



PASSIVITY BASED DESIGN OF SYNCHRONIZING SYSTEMS

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A particular problem of controlled synchronization of nonlinear systems is considered. Minimum conditions for producing such synchronization are presented and it is shown that these conditions also provide the *feedback passivity* of the overall system. It is also demonstrated that the conditions of feedback passivity allow one to design an adaptive synchronizing control law which ensures global synchronization in the cases when the systems to be synchronized have different parameters. Additionally it is demonstrated that the approach presented for the design of synchronizing systems allows one to cope with external disturbances in order to protect the systems from loss of synchrony.

1. Introduction

Synchronization of dynamical systems is a phenomenon which is currently the focus of attention of researchers in various fields of science. It is widely known in mechanical [Blekhman, 1971, 1988] and electrical systems (see e.g. [Wu & Chua, 1993; Kocarev *et al.*, 1992] and references therein). A particular point of interest in the problem of synchronization of dynamical systems was highlighted in the paper [Pecora & Carroll, 1990] where it was reported that two chaotic systems being interconnected may synchronize. At first sight this result seems amazing because chaos implies instability while synchronization is a property of stability. This contradiction can easily be resolved: solutions of the overall system may be asymptotically stable with respect to some variables and locally unstable with respect to others. This simple observation was extensively used in [Wu & Chua, 1994] in order to give a unified framework for the design and analysis of the synchronizing systems. Nevertheless

in applications there are demands for more sophisticated tools for the design of synchronizing systems. In this paper we present such an approach based on the concepts of dissipativity in the sense of Willems [1972] and passivity in the sense of Hill–Moylan [Moylan, 1974; Hill & Moylan, 1976; Byrnes *et al.*, 1991].

Particular problems of controlled synchronization are widely used by control theorists mostly for the problems of observation (reconstruction of the whole state vector of a dynamical system) and model reference control (this problem arises when one wants to make the dynamics of the controlled system close to the dynamics of a given system which in this case is referred to as a *reference model*). The connection between the problem of controlled synchronization on one hand and conventional problems of observation and model reference control on the other, was recently reported [Nijmeijer & Mareels, 1996; Morgül & Solak, 1996; Fradkov, 1995]. These results allow one to

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systematically design a wide class of synchronizing systems but there is still a gap between the obtained experimental and numerical data and the mathematical results. Indeed the problems of observation and model reference control, being dual to each other, are very close to the case of the so-called “master-slave” synchronization which means that there is a unidirectional interconnection between the synchronizing systems. At the same time the case of mutual interconnections is a more general one and it may be considered within the same framework. An additional mathematical difficulty is that the solution of the coupled system in this case does not exist *a priori* on the infinite time interval and as we will see later, some additional technical assumptions allow one to establish that the overall system consisting of mutually interconnected systems has a bounded solution and therefore it exists for all time instances. The rest of the mathematical problems are conventional for control theorists and involve stability analysis in the spirit of the direct Lyapunov method.

Throughout the paper we will use the concepts of dissipativity and passivity. These concepts are very important in modern control theory. They involve consideration of some scalar functions (storage functions) associated with the controlled system (i.e. the system with input and output). Such scalar functions play the same role as Lyapunov functions in classical stability theory. It is well known that the direct Lyapunov method can be successfully applied to the study of stability of sets (see e.g. [Lin *et al.*, 1996]). On the other hand some particular cases of the synchronization phenomenon can be considered by use of the concept of set stability and it gives a hint that some known results of output stabilization of minimum-phase systems (see e.g. [Hill & Moylan, 1976]) can be reformulated to cover the case of controlled synchronization. This attempt is made in the present paper and it turns out that the sufficient conditions of existence of the synchronizing output feedback are the conditions of existence of the output feedback which makes the overall system passive. This observation allows one to use concepts of passivity and minimum-phasesness for the systematic design of oscillatory synchronizing systems.

At the same time external disturbances affecting the plant may cause instability and loss of synchronism. To cope with the external disturbance in passive systems, one can implement a high-gain controller which renders all solutions of the overall

system bounded with some prespecified bounds or variable-structure controller which ensures asymptotic stability. These two controllers have some drawbacks since the high gain is difficult to realize in practice. On the other hand in practice we often face the situation when the external disturbance is a periodic or almost periodic function of time which can be represented as a finite trigonometric series with unknown coefficients. It means that we know *a priori* only a class of possible disturbances but not their exact waveforms. Nevertheless as it will be shown this assumption together with some kind of passivity (passifiability) property of the plant ensures the existence of a smooth adaptive compensator for such a disturbance. It is important to note that this compensator depends only on output on-line measurements, i.e. the problem of external disturbance compensation can be solved with incomplete information about the system state.

The main contribution of this paper is that it shows that various problems of synchronization, adaptive synchronization and disturbance attenuation can be solved simultaneously in a unified framework based on the use of the passivity property. This framework allows one to solve the problem of synchronizing feedback design with incomplete measurements of the system states and for mutually coupled systems. These key features distinguish the present approach from the existing ones.

Throughout the paper we will use some ideas of [Fradkov, 1995]. The main novelty of the paper is that the result of [Fradkov, 1995] is generalized to nonlinear systems of a more general structure than in [Fradkov, 1995] where the case of Lur’e systems was considered. Additionally in this paper the problem of synchronization in presence of disturbances is considered.

The paper is organized as follows. First we introduce some useful definitions and pose the problem of synchronization of dynamical systems as a control problem. In Sec. 4 a specific case of the posed problem is solved under assumptions which ensure passivity of the overall system. Some aspects of synchronization of several systems with multiple interconnections are discussed in Sec. 5. In Sec. 6 we discuss the connection between the convergency property of dynamical systems (concept of convergent systems was introduced by V. A. Pliss and generalized by B. P. Demidovich) and the synchronization phenomenon. Further in Sec. 7 some results ensuring adaptive synchronization are derived based on the concepts of passivity and speed

gradient. In Sec. 8 we present some approaches that can help to retain the synchrony if there are unmeasurable disturbances affecting the systems. Some theoretical results obtained in this paper are illustrated by the example of synchronization of Lorenz systems.

2. Nomenclature and Preliminaries

An $\mathcal{L}_p$ norm, $1 \leq p \leq \infty$ is denoted as $\|\cdot\|_p$. The Euclidean norm in $\mathbb{R}^n$ is denoted simply as $|\cdot|$, $|x|^2 = x^\top x$, where $^\top$ stands for the transpose operation. More generally, we will study notions relative to nonempty subsets $\mathcal{A}$, of $\mathbb{R}^n$, $0 \in \mathcal{A}$; for such a set $\mathcal{A}$, $|x|_{\mathcal{A}} = \text{dist}(x, \mathcal{A}) = \inf_{\eta \in \mathcal{A}} \text{dist}(x, \eta)$ denotes the point-to-set distance from $x \in \mathbb{R}^n$ to $\mathcal{A}$.

A function $V : X \rightarrow \mathbb{R}^+$ defined on a subset X of $\mathbb{R}^n$, $0 \in X$ is *positive definite* if $V(x) > 0$ for all $x \in X \setminus \{0\}$ and $V(0) = 0$. It is *radially unbounded* if $V(x) \rightarrow \infty$ as $|x| \rightarrow \infty$. A nonnegative function $V : X \rightarrow \mathbb{R}^+$ is said to be *positive definite with respect to the set $\mathcal{A}$* if $V(x) > 0$ for all $x \in X \setminus \mathcal{A}$ and $V(x) = 0$ for all $x \in \mathcal{A}$. It is *proper with respect to $\mathcal{A}$* if boundedness of $V(x)$ implies boundedness of $|x|_{\mathcal{A}}$. In the paper we use the following notations: let $V : \mathbb{R}^n \rightarrow \mathbb{R}^1$ be a smooth enough vector function then $\nabla V(x)$ stands for the vector (in $\mathbb{R}^n$) of its first partial derivatives calculated at the point x : $(\nabla V(x))^\top = \partial V(x)/\partial x$, for example $(\nabla V(x))^\top f(x)$ stands for the Lie derivative of the function V with respect to the vector field f calculated at the point x . More generally if $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ then $\partial f(x)/\partial x$ is an $m \times n$ matrix composed of the partial derivatives $\partial f_i(x)/\partial x_j$, $i = 1, \dots, m$, $j = 1, \dots, n$.

Consider the nonlinear time-invariant affine in the control system:

$$\begin{cases} \dot{x} = f(x) + g(x)u \\ y = h(x) \end{cases} \quad (1)$$

where $x(t) \in \mathbb{R}^n$ is the state, $u(t) \in \mathbb{R}^m$ is the input which is assumed to be a continuous and bounded function of time: $u \in \mathcal{C}^0 \cap \mathcal{L}_\infty$, $y(t) \in \mathbb{R}^l$ is the output; $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and the columns of the matrix $g : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ are smooth vector fields, $f(0) = 0$ and $h : \mathbb{R}^n \rightarrow \mathbb{R}^l$ is a smooth mapping. Associated with the system (1) consider a real-valued defined on $\mathbb{R}^l \times \mathbb{R}^m$ function w called the *supply rate*. We will assume that this function is well defined on any compact subset of the set $\mathbb{R}^l \times \mathbb{R}^m$.

Definition 1. The system (1) with supply rate w is said to be $\mathcal{C}^r$ -dissipative in the sense of Willems if there exists a $\mathcal{C}^r$ -smooth, $r \geq 0$ nonnegative function $V : \mathbb{R}^n \rightarrow \mathbb{R}^+$, called the *storage function*, such that the following *dissipation inequality* holds:

$$V(x(t)) - V(x(0)) \leq \int_0^t w(y(s), u(s)) ds \quad (2)$$

for all $u \in \mathcal{C}^0 \cap \mathcal{L}_\infty$, $x(0) \in \mathbb{R}^n$, $0 \leq t < T_{u, x_0}$, where T_{u, x_0} is the upper time limit for which the solution corresponding to the input u and initial conditions $x(0) = x_0$ exists.

This definition has a clear physical interpretation if the storage function is understood as the total energy of the system. Then the left hand side of this inequality is the increment of the energy up to time t and the dissipation inequality means that this increment should not exceed integral of the supply rate.

Throughout the paper we are interested in the particular case when the supply rate is linear in control, that is $w(y, u) = q(y)^\top u$ for some function $q : \mathbb{R}^l \rightarrow \mathbb{R}^m$. In this case mapping $q \circ h$ defines a new output of the same dimension as input u . Motivated by this observation we will consider the case of square systems: $l = m$.

Definition 2. The system (1) is $\mathcal{C}^r$ -passive if it is $\mathcal{C}^r$ -dissipative with supply rate $w(y, u) = y^\top u$ and the storage function V satisfies $V(0) = 0$.

When studying the passivity property for the systems with differentiable storage functions it is possible to give an alternative definition of a passive system. This definition can be considered as the infinitesimal version of the Definition 2. System (1) is said to be $\mathcal{C}^r$ -passive, $r \geq 1$ if there exists a nonnegative function $V : \mathbb{R}^n \rightarrow \mathbb{R}^+$, $V(0) = 0$ such that

$$\begin{aligned} (\nabla V)^\top f &\leq 0 \\ (\nabla V)^\top g &= h^\top \end{aligned}$$

The fact that these two definitions are equivalent can be proved by the use of the *semigroup property* of dynamical systems and is referred to as the *nonlinear Kalman–Yakubovich–Popov Lemma* [Hill & Moylan, 1976; Byrnes *et al.*, 1991].

The above definitions are classical in the nonlinear control theory but in this paper we

also need some weakened version of the passivity property:

Definition 3. The system (1) is called $\mathcal{C}^r$ -semipassive if there exist a $\mathcal{C}^r$ -smooth, $r \geq 0$ non negative function $V : \mathbb{R}^n \rightarrow \mathbb{R}^+$ and a function $H : \mathbb{R}^n \rightarrow \mathbb{R}^1$ such that for any initial conditions $x(0)$ and any admissible input $u \in \mathcal{C}^0 \cap \mathcal{L}_\infty$ the following *dissipation inequality*

$$V(x(t)) - V(x(0)) \leq \int_0^t (y(s)^\top u(s) - H(x(s))) ds \quad (3)$$

holds for all $0 \leq t < T_{u, x_0}$, where the function H is nonnegative outside some ball:

$$\exists \rho > 0 \quad |x| \geq \rho \implies H(x) \geq 0.$$

The most useful property of the semipassive systems is that being linearly interconnected they possess bounded solutions. More precisely we formulate the following result:

Consider the following systems:

$$\begin{cases} \dot{x}_1 = f_1(x_1) + g_1(x_1)u_1 \\ y_1 = h_1(x_1) \end{cases} \quad (4)$$

$$\begin{cases} \dot{x}_2 = f_2(x_2) + g_2(x_2)u_2 \\ y_2 = h_2(x_2) \end{cases} \quad (5)$$

where $x_1(t) \in \mathbb{R}^{n_1}$, $x_2(t) \in \mathbb{R}^{n_2}$ are the states, $u_1(t) \in \mathbb{R}^m$, $u_2(t) \in \mathbb{R}^m$ are the inputs, $y_1(t) \in \mathbb{R}^m$, $y_2(t) \in \mathbb{R}^m$ are the outputs and the functions $f_1, f_2, g_1, g_2, h_1, h_2$ are smooth enough to ensure existence, at least on some time interval, of the solutions to the system closed by the following feedback:

$$u_1 = -\gamma(y_1 - y_2), \quad u_2 = -\gamma(y_2 - y_1) \quad (6)$$

where γ is some positive number.¹

Lemma 1. Suppose that the systems (4), (5) are $\mathcal{C}^r$ -semipassive ($r \geq 0$) with radially unbounded storage functions V_1 and V_2 . Then all solutions of the coupled system (4)–(6) exist on the infinite time interval and bounded.

Proof. Rewrite the dissipation inequality for each system

$$\begin{aligned} & V_1(x_1(t)) - V_1(x_1(0)) \\ & \leq \int_0^t (y_1(s)^\top u_1(s) - H_1(x_1(s))) ds \end{aligned} \quad (7)$$

$$\begin{aligned} & V_2(x_2(t)) - V_2(x_2(0)) \\ & \leq \int_0^t (y_2(s)^\top u_2(s) - H_2(x_2(s))) ds \end{aligned} \quad (8)$$

Adding inequalities (7) and (8) and substituting (6), we obtain

$$\begin{aligned} & W(x(t)) - W(x(0)) \\ & \leq \int_0^t (-y_1(s)^\top \gamma y_1(s) + y_1(s)^\top \gamma y_2(s) \\ & \quad + y_2(s)^\top \gamma y_1(s) - y_2(s)^\top \gamma y_2(s) \\ & \quad - H_1(x_1(s)) - H_2(x_2(s))) ds \end{aligned}$$

where the nonnegative radially unbounded function W is defined as

$$W(x) = V_1(x_1) + V_2(x_2), \quad x = \text{col}(x_1, x_2).$$

Notice that $-y_1^\top y_1 + 2y_1^\top y_2 - y_2^\top y_2 \leq 0$.

Now consider the following compact set $\Omega = \{x : |x_1| \leq \rho_1, |x_2| \leq \rho_2\}$, where the numbers ρ_1, ρ_2 are from the definition of semipassive systems. Consider the following scalar function $W_1 : \mathbb{R}^{n_1+n_2} \rightarrow \mathbb{R}^+$:

$$W_1(x) = \begin{cases} 0, & \text{if } x \in \Omega \\ W(x), & \text{otherwise} \end{cases}$$

Clearly, the function W_1 is bounded on any compact set and it is radially unbounded. From the dissipation inequality we readily obtain that

$$W_1(x(t)) - W_1(x(0)) \leq 0.$$

From the last inequality one can deduce that $W_1(x(t))$ is bounded for all $t \geq 0$ and therefore all solutions exist on the infinite time interval and are bounded. ■

In the sequel we will also use the concepts of strict passivity and exponential passivity with respect to sets.

Definition 4. A $\mathcal{C}^r$ -passive system (1) such that $h(x) = 0$ if $x \in \mathcal{A}$ is said to be $\mathcal{C}^r$ -strictly passive with respect to $\mathcal{A}$, $r \geq 0$ if there exist a

¹The result holds true if γ is some positive definite $m \times m$ matrix.

$\mathcal{C}^r$ -smooth, $r \geq 0$, nonnegative function $V : \mathbb{R}^n \rightarrow \mathbb{R}^+$ and a positive definite with respect to $\mathcal{A}$ function $S : \mathbb{R}^n \rightarrow \mathbb{R}^+$, such that $V(x) = 0$ for all $x \in \mathcal{A}$ and the following relation

$$\begin{aligned} V(x(t)) - V(x(0)) \\ = \int_0^t (y(s)^\top u(s) - S(x(s))) ds \end{aligned} \quad (9)$$

holds for all $0 \leq t < T_{u,x_0}$ and for all $u \in \mathcal{C}^0 \cap \mathcal{L}_\infty$.

Definition 5. The system (1) is $\mathcal{C}^r$ -exponentially passive with respect to $\mathcal{A}$, $r \geq 0$ if it is $\mathcal{C}^r$ -strictly passive with respect to $\mathcal{A}$ and there exist positive numbers $\alpha_1, \alpha_2, \alpha_3$ such that the following inequalities hold:

$$\alpha_1 |x|_{\mathcal{A}}^2 \leq V(x) \leq \alpha_2 |x|_{\mathcal{A}}^2, \quad \alpha_3 |x|_{\mathcal{A}}^2 \leq S(x) \quad (10)$$

for any solution $x(t)$ to (1).

For comprehensive understanding of the results of this paper the reader should be familiar with the notions of relative degree, zero dynamics and minimum-phasesness and we recommend the excellent paper by Byrnes, Isidori and Willems [1991] where the necessary background is presented in very convenient form.

3. Synchronization as a Control Problem

Synchronization of dynamical systems is a commonly known phenomenon. Synchronization in its most general interpretation means correlated or corresponding in time behavior of two or more processes. Below we will formulate a definition of the synchronization phenomenon. This definition is a particular case of the definition given in [Blekhman *et al.*, 1997].

Consider k dynamical systems described by the k interconnected systems of ordinary differential equations:

$$S_i : \dot{x}_i = F_i(x_1, x_2, \dots, x_k, t), \quad i = 1, \dots, k \quad (11)$$

where $F_i : \mathbb{R}^{n_1} \times \dots \times \mathbb{R}^{n_k} \times \mathbb{R}^+ \rightarrow \mathbb{R}^{n_i}$. In this paper, for the sake of simplicity we will give a coordinate dependent definition of synchronization. The more general coordinate-free definition of the problem of controlled synchronization involves the coordinate-free definition of dynamical systems and can be

found for example in [Blekhman *et al.*, 1997]. Associated with the set of systems (11) consider some time-dependent functional Q (one-parametric family of the functionals Q) defined on the solutions $x_i(\cdot)$ to these systems: $Q : \mathcal{X}_1 \times \dots \times \mathcal{X}_k \times \mathbb{R}^+ \rightarrow \mathbb{R}$, where $\mathcal{X}_i \subset \{x_i : \mathbb{R}^+ \rightarrow \mathbb{R}^{n_i}\}$. We will say that solutions to the systems S_i are synchronized if the value of the functional Q is identically zero for these solutions for all $t \geq 0$. More precisely, to define synchronization we also need to define some different parametrizations of time. For any $\tau \in \mathbb{R}^+$ we define σ_τ as the *shift operator*, i.e. $\sigma_\tau : \mathcal{X}_i \rightarrow \mathcal{X}_i$ is given as $(\sigma_\tau x)(t) = x(t + \tau)$ for all $x \in \mathcal{X}_i$ and all $t \in \mathbb{R}^+$. We are now prepared to define synchronization.

Definition 6. Solutions $x_1(t), \dots, x_k(t)$ of the systems $S_1, \dots, S_k$ with initial conditions $x_1(0), \dots, x_k(0)$ are called *synchronized with respect to the functional Q* if there exist $\tau_1, \dots, \tau_k \in \mathbb{R}^+$ such that

$$Q(\sigma_{\tau_1} x_1(\cdot), \dots, \sigma_{\tau_k} x_k(\cdot), t) \equiv 0, \quad (12)$$

for all $t \in \mathbb{R}^+$.

The solutions $x_1(t), \dots, x_k(t)$ of the systems $S_1, \dots, S_k$ with initial conditions $x_1(0), \dots, x_k(0)$ are *asymptotically synchronized with respect to the functional Q* , if for some $\tau_1, \dots, \tau_k \in \mathbb{R}^+$

$$\lim_{t \rightarrow \infty} (Q(\sigma_{\tau_1} x_1(\cdot), \dots, \sigma_{\tau_k} x_k(\cdot), t)) = 0 \quad (13)$$

Remark 1. If one allows for the operator σ_τ to depend on time, then the phase shifts between the synchronous variables may not be constant. The case when these shifts tend to some constant values corresponds to the case of *asymptotic phases*.

In the case of arbitrary initial conditions $x_i(0)$ this phenomenon will be referred to as the *global synchronization problem*, while for certain admissible initial conditions we will talk about *conditional synchronization*.

If the equations of the interconnected systems (11) are given then the problem of synchronization is one of analysis. A more interesting case is when one should find a control law to ensure synchronization. Suppose that the systems $S_1, \dots, S_k$ can be controlled:

$$S_i : \dot{x}_i = F_i(x_1, x_2, \dots, x_k, u, t), \quad i = 1, \dots, k \quad (14)$$

where $F_i : \mathbb{R}^{n_1} \times \dots \times \mathbb{R}^{n_k} \times \mathbb{R}^m \times \mathbb{R}^+ \rightarrow \mathbb{R}^{n_i}$ and $u(t) \in \mathbb{R}^m$ is the control action.

The problem of output feedback synchronization can be posed as follows: find controller equations

$$u(t) = \mathcal{U}(y_1(s), \dots, y_k(s), t, 0 \leq s \leq t) \quad (15)$$

where $y_i(t) \in \mathbb{R}^{l_i}, i = 1, \dots, k$ are outputs of each system: $y_i = h_i(x_i, u, t)$ and $\mathcal{U}$ is some operator such that the goal (13) in the system (14), (15) is achieved. If the controller (15) ensures the goal (13) for arbitrary initial conditions the feedback (15) will be referred to as a *globally synchronizing feedback*. Similarly in the case of semiglobal stabilization, it is possible to pose the problem of design of *semiglobally synchronizing feedback*.

Definition 7. The systems (14) are *semiglobally asymptotically synchronizable with respect to the functional Q by output feedback* if for any compact set of initial conditions $\Omega \subset \mathbb{R}^{n_1} \times \dots \times \mathbb{R}^{n_k}$ there exists an output feedback of the form (15) perhaps depending on Ω , such that in the closed-loop system (14), (15) the goal (13) is achieved for all initial conditions from the set Ω .

In what follows, for the sake of simplicity we consider the case of synchronization of two dynamical systems ($k = 2$) of the same order ($n_1 = n_2 = n$). A special case of the posed problem arises when the functional Q is given by

$$Q(x_1(\cdot), x_2(\cdot), t) = |x_1(t) - x_2(t)|,$$

then the problem of synchronization analysis and synthesis can be solved by the use of the concept of stability of sets. Indeed, if all solutions $x_1(t)$ and $x_2(t)$ are bounded functions of time and additionally $Q(x_1(\cdot), x_2(\cdot), t) \rightarrow 0$ as $t \rightarrow \infty$ then the overall system generally possesses an asymptotically stable set which is a compact subset of the set defined as $\{\text{col}(x_1, x_2) \in \mathbb{R}^{2n} : x_1 = x_2\}$. Moreover in this particular case this stable set has a simple structure: It consists of the closure of trajectory $\{\bar{x}\}$ where $\bar{x}$ is the asymptotically stable solution (in the sense that $|x_1(t) - \bar{x}(t)| \rightarrow 0, |x_2(t) - \bar{x}(t)| \rightarrow 0$ and $\bar{x}(t)$ is unique and bounded for $-\infty < t < \infty$ Lyapunov stable solution) which is referred to as a *synchronous solution*. In more general cases the limit set which corresponds to the synchronous mode may consist of closures of trajectories which admit solutions with the asymptotic phase.

4. Synchronization of Linearly Coupled Systems

In this section we will apply some concepts discussed above in order to investigate occurrence of the synchronization in some particular synchronization schemes. More precisely we will study synchronization of two identical systems. Consider the following system:

$$\begin{cases} \dot{x}_1 = f(x_1) + g_1(x_1)u_1 \\ \dot{x}_2 = f(x_2) + g_2(x_2)u_2 \end{cases} \quad (16)$$

where $x_1(t) \in \mathbb{R}^n$ and $x_2(t) \in \mathbb{R}^n$ are the states variables, $u_1(t) \in \mathbb{R}^m, u_2(t) \in \mathbb{R}^m$ are the control inputs and $y_1(t) \in \mathbb{R}^m$ and $y_2(t) \in \mathbb{R}^m$ are the outputs of the first and second subsystems given by

$$y_1 = h(x_1), \quad y_2 = h(x_2) \quad (17)$$

We will assume that f, g_1, g_2 and h are smooth enough to ensure existence of a unique solution of (16) with continuous in t inputs u_1, u_2 at least on some time interval. The problem of synchronization is to find an appropriate feedback law such that $|x_1(t) - x_2(t)| \rightarrow 0$ as $t \rightarrow \infty$ for all initial conditions $x_1(0), x_2(0)$.

In the study of this type of synchronization we will assume also that each system to synchronize has relative degree $(1, \dots, 1)^\top$. Recall that this assumption means that matrices $(\nabla h(x))^\top g_1(x)$ and $(\nabla h(x))^\top g_2(x)$ are nonsingular in the neighborhood of origin [Isidori, 1989]. If additionally the distribution spanned by the columns of the matrix $g_1(x)$ (the same for $g_2(x)$) is involutive then it is possible to find coordinate transformations $z_i = \Phi_i(x_i), z_i \in \mathbb{R}^{n-m}, i = 1, 2$, locally defined near origin such that in the new coordinates the system is represented in *Isidori's normal form*.

$$\begin{cases} \dot{z}_1 = q(z_1, y_1) \\ \dot{y}_1 = a(z_1, y_1) + b_1(z_1, y_1)u_1 \\ \dot{z}_2 = q(z_2, y_2) \\ \dot{y}_2 = a(z_2, y_2) + b_2(z_2, y_2)u_2 \end{cases} \quad (18)$$

Moreover we will assume that such a coordinate transformation is globally defined, that is the system (18) is globally diffeomorphic to the original system. This assumption is satisfied if all columns of the matrices $g_i(x)((\nabla h(x))^\top g_i(x))^{-1}$ are complete vector fields and the matrices $(\nabla h(x))^\top g_i(x)$

are nonsingular for all x , $i = 1, 2$ [Byrnes & Isidori, 1991].

The problem is to find an appropriate control algorithm as an output static feedback to ensure the goal of synchronization. We will look for the conditions which ensure synchronization by the simplest control algorithm:

$$u_1 = -\gamma(y_1 - y_2), \quad u_2 = -\gamma(y_2 - y_1) \quad (19)$$

where $\gamma \in \mathbb{R}^1$ is the synchronization gain, also referred to in the literature as a *coupling constant*. As we will see, under certain conditions if γ exceeds some threshold value then the synchronization occurs for all initial conditions from a given compact set. Imposing the constraint $y_1 - y_2 = 0$ one can easily find dynamics of the constrained system (18):

$$\begin{cases} \dot{z}_1 = q(z_1, y_1) \\ \dot{z}_2 = q(z_2, y_1) \\ \dot{y}_1 = a(z_1, y_1) \end{cases} \quad (20)$$

The system (20) provides the *zero dynamics* of system (18).

Now we are in position to formulate the following result which establishes sufficient conditions of the semiglobal synchronization:

Theorem 1. *Assume that*

- A1. *The functions q , a , b_1 , b_2 are continuous and locally Lipschitz.*
- A2. *System*

$$\begin{cases} \dot{z}_i = q(z_i, y_i) \\ \dot{y}_i = a(z_i, y_i) + b_i(z_i, y_i)u_i \end{cases} \quad (21)$$

$i = 1, 2$, is $\mathcal{C}^0$ -semipassive with respect to the input u_i and output y_i with the radially unbounded storage function $V_i : \mathbb{R}^n \rightarrow \mathbb{R}^+$.

- A3. *There exists a $\mathcal{C}^2$ -smooth positive definite function $V_0 : \mathbb{R}^{n-m} \rightarrow \mathbb{R}^+$ and positive number α such that the following inequality is satisfied*

$$\begin{aligned} & (\nabla V_0(z_1 - z_2))^\top (q(z_1, y_1) - q(z_2, y_1)) \\ & \leq -\alpha |z_1 - z_2|^2. \end{aligned}$$

for all $z_1, z_2 \in \mathbb{R}^{n-m}$, $y_1 \in \mathbb{R}^m$.

- A4. *The matrix $b_1(z_1, y_1) + b_2(z_2, y_2)$ is positive definite:*

$$b_1(z_1, y_1) + b_2(z_2, y_2) > 2\beta I_m, \quad \beta > 0.$$

Then for any initial conditions $z_1(0), z_2(0), y_1(0), y_2(0)$ there exists $\bar{\gamma}$ such that for all $\gamma > \bar{\gamma}$ all solutions $z_1(t), z_2(t), y_1(t), y_2(t)$ are bounded for all $t \geq 0$ and the goal of synchronization is achieved.

Proof. By virtue of Lemma 1 all solutions of (18) are bounded and exist for any $0 \leq t < \infty$: $|z_1(t)| \leq \mathcal{B}_z$, $|z_2(t)| \leq \mathcal{B}_z$, $|y_1(t)| \leq \mathcal{B}_y$, $|y_2(t)| \leq \mathcal{B}_y$ for all $t \geq 0$ and some positive numbers $\mathcal{B}_z, \mathcal{B}_y$ which may depend on the initial conditions. It is important that all solutions are bounded by a constant which does not depend on the synchronization gain γ (see proof of Lemma 1).

Then consider the following Lyapunov function candidate:

$$\begin{aligned} V(z_1 - z_2, y_1 - y_2) \\ = V_0(z_1 - z_2) + \frac{1}{2}(y_1 - y_2)^\top (y_1 - y_2). \end{aligned}$$

Its time derivative with respect to the system (18) satisfies:

$$\begin{aligned} \dot{V} &= (\nabla V_0(z_1 - z_2))^\top (q(z_1, y_1) - q(z_2, y_2)) \\ &+ (y_1 - y_2)^\top (a(z_1, y_1) - a(z_2, y_2)) \\ &- \gamma(y_1 - y_2)^\top (b_1(z_1, y_1) \\ &+ b_2(z_2, y_2))(y_1 - y_2) \\ \dot{V} &= (\nabla V_0(z_1 - z_2))^\top (q(z_1, y_1) - q(z_2, y_1)) \\ &+ (\nabla V_0(z_1 - z_2))^\top (q(z_2, y_1) - q(z_2, y_2)) \\ &+ (y_1 - y_2)^\top (a(z_1, y_1) - a(z_2, y_2)) \\ &- \gamma(y_1 - y_2)^\top ((b_1(z_1, y_1) \\ &+ b_2(z_2, y_2))(y_1 - y_2)). \end{aligned}$$

We now aim to determine number $\bar{\gamma} > 0$ such that $\dot{V}$ is nonpositive on the set $|z_1| \leq \mathcal{B}_z$, $|z_2| \leq \mathcal{B}_z$, $|y_1| \leq \mathcal{B}_y$, $|y_2| \leq \mathcal{B}_y$ as $\gamma > \bar{\gamma}$. Using smoothness of the right hand side of (18), $\mathcal{C}^2$ -smoothness of V_0 and boundedness of all solutions $z_1(t), z_2(t), y_1(t), y_2(t)$ we have

$$\begin{aligned} |q(z_2, y_1) - q(z_2, y_2)| &\leq C_1 |y_1 - y_2| \\ |a(z_1, y_1) - a(z_2, y_2)| &\leq C_2 |y_1 - y_2| \\ &+ C_3 |z_1 - z_2| \\ |\nabla V_0(z_1 - z_2)| &\leq C_4 |z_1 - z_2| \end{aligned} \quad (22)$$

for some positive constants C_1, C_2, C_3, C_4 and all $|z_1| \leq \mathcal{B}_z, |z_2| \leq \mathcal{B}_z, |y_1| \leq \mathcal{B}_y, |y_2| \leq \mathcal{B}_y$. Taking into account Assumption A3, we have

$$\begin{aligned} \dot{V} \leq & -\alpha|z_1 - z_2|^2 \\ & + (C_4C_1 + C_3)|z_1 - z_2| \cdot |y_1 - y_2| \\ & - (2\gamma\beta - C_2)|y_1 - y_2|^2. \end{aligned}$$

Hence, choosing $\bar{\gamma}$ to satisfy the inequality

$$\bar{\gamma} \geq \frac{1}{2\beta} \left(C_2 + \frac{(C_4C_1 + C_3)^2}{4\alpha} \right)$$

we just obtain that $\dot{V}$ is nonpositive, that is V is a nonincreasing function of time and therefore it is bounded for any solution to the whole system. If $\gamma > \bar{\gamma}$ then there exists $\varepsilon > 0$ such that

$$\dot{V} \leq -\varepsilon(|z_1 - z_2|^2 + |y_1 - y_2|^2) \quad (23)$$

Integrating this inequality over $[0, \infty)$ (recall that according to Lemma 1 all solutions exist on the infinite time interval) yields:

$$\begin{aligned} V(z_1(0) - z_2(0), y_1(0) - y_2(0)) \\ \geq \varepsilon \int_0^\infty (|z_1(s) - z_2(s)|^2 \\ + |y_1(s) - y_2(s)|^2) ds \end{aligned} \quad (24)$$

That is the right hand side of this inequality is bounded, i.e. the integral exists and is finite. Consequently, all solutions of the whole system are bounded, the right hand side of the system equations is locally Lipschitz continuous, therefore the right hand side of the closed system is bounded for any solutions, that is all solutions are uniformly continuous in t . Therefore $|z_1(t) - z_2(t)|^2 + |y_1(t) - y_2(t)|^2$ is uniformly continuous in t as well. The result now follows from the Barbalat lemma:

Lemma 2 (Barbalat). [Popov, 1973] Consider the function $\psi : \mathbb{R}^1 \rightarrow \mathbb{R}^1$. If ψ is uniformly continuous and $\lim_{t \rightarrow \infty} \int_0^\infty \psi(s) ds$ exists and is finite then

$$\lim_{t \rightarrow \infty} \psi(t) = 0$$

Remark 2. Since Assumption A2 implies boundedness of all solutions of the overall system it is sufficient to require that Assumption A3 is valid only

on some compact set $\{z_1, z_2, y_1 : |z_1| \leq \mathcal{B}_z, |z_2| \leq \mathcal{B}_z, |y_1| \leq \mathcal{B}_y\}$.

Since the bounds $\mathcal{B}_z, \mathcal{B}_y$ may depend on the initial conditions Theorem 1 gives semiglobal result. In the next section it will be shown that under some additional conditions (property of strict semipassivity) these bounds become the ultimate bounds, i.e. they do not depend on the initial conditions and therefore the feedback (19) ensures *global synchronization*.

Note that Assumption A2 requires that all the systems to synchronize are semipassive. If this is the case then the synchronizing feedback can be found in the form (19). If one looks for the conditions which ensure existence of the synchronizing feedback of a more general class, e.g. in the class of output feedbacks then it is sufficient to require merely that the system (21) is *equivalent via output feedback* to a semipassive system (equivalence via feedback to a passive system is referred also to as *feedback passivity* or simply *passifiability*). Conditions which ensure such equivalence can be derived similarly to [Fradkov & Hill, 1993] where there were found conditions of the equivalence of a given system to an exponentially passive system via output static feedback law. According to [Fradkov & Hill, 1993] one can find that sufficient condition for semiglobal equivalence via static output feedback of the system (21) to a semipassive system is that it is globally hyperbolically minimum-phase and the matrix $b_i(z_i, y_i)$ admits the following factorization $b_i(z_i, y_i) = b'_i(z_i)b''_i(y_i)$, where $b''_i(y_i)$ is nonsingular and $b'_i(z_i)$ is positive definite. Notice also that the condition of global hyperbolic minimumphases of the system (21) directly follows from the Assumption A3 if, additionally the function V_0 satisfies quadratic type inequalities on any compact set.

Remark 3. It is also worth to mention that the assumption about existence of a globally defined normal form can be weakened. Indeed, as it is seen from the proof of Theorem 1 we do not need that each matrix $b_i(z_i, y_i)$ is nonsingular. In fact all we need is Assumption A4. This assumption can be satisfied for example if $b_1(z, y) \equiv 0$ and $b_2(z, y)$ is positive definite. In this case instead of Assumption A2 one may require that the first system (the “master system”, which is not affected by the control) is Lagrange stable or, more mildly, that it has solutions well defined for all time instances $0 \leq t < \infty$ at least for admissible initial conditions.

5. Synchronization of Systems with Multiple Interconnections

In this section we will consider some aspects of synchronization of several systems with multiple interconnections. Based on the result of the previous section it is not surprising that under some conditions the systems may synchronize if the coupling between them exceeds some threshold value. A more precise statement depends on the way the systems interact and here we will briefly discuss the case when every system is interconnected to each other. Let us consider the case of synchronization by linear symmetric coupling of k identical systems. As before we will assume that each system to synchronize is globally diffeomorphic to Isidori's normal form and additionally suppose that "high-frequency gains" of each system $b_i(z_i, y_i), i = 1, \dots, k$ are equal to each other and constant: $b_i(z_i, y_i) = b_j(z_j, y_j) \equiv B$, where $B = B^\top > 0$ is some positive definite $m \times m$ matrix. Under these assumptions each system can be described by the following equation:

$$\begin{cases} \dot{z}_i = q(z_i, y_i) \\ \dot{y}_i = a(z_i, y_i) - B \left(\sum_{s \neq i}^k \gamma(y_i - y_s) \right) \end{cases}$$

or

$$\begin{cases} \dot{z}_i = q(z_i, y_i) \\ \dot{y}_i = a(z_i, y_i) + \gamma B \sum_{s=1}^k y_s - \gamma B k y_i \end{cases} \quad (25)$$

where $\gamma > 0$ is the coupling constant.

As before suppose that the system

$$\begin{cases} \dot{z} = q(z, y) \\ \dot{y} = a(z, y) + Bu \end{cases} \quad (26)$$

is semipassive with radially unbounded continuous storage function $V_1 : \mathbb{R}^n \rightarrow \mathbb{R}^+$ and there exists a twice differentiable positive definite scalar function $V_0 : \mathbb{R}^{n-m} \rightarrow \mathbb{R}^+$ which satisfies

$$(\nabla V_0(z_i - z_j))^\top (q(z_i, y_i) - q(z_j, y_i)) \leq -\alpha |z_i - z_j|^2$$

for any $i, j \leq k$ and some positive α . Then by virtue of Lemma 1 all solutions of the whole system are bounded and similarly to the proof of Theorem 1 examining the following Lyapunov function

candidate

$$V(\bar{z}, \bar{y}) = \sum_{i=1}^k \sum_{j=i+1}^k \left(V_0(z_i - z_j) + \frac{1}{2} (y_i - y_j)^\top (y_i - y_j) \right) \quad (27)$$

where $\bar{z} = \text{col}(z_1, \dots, z_k), \bar{y} = \text{col}(y_1, \dots, y_k)$ one can conclude that the following goal of synchronization

$$\lim_{t \rightarrow \infty} \sum_{i=1}^k \sum_{j=i+1}^k (|z_i(t) - z_j(t)| + |y_i(t) - y_j(t)|) = 0 \quad (28)$$

is achieved as long as the synchronization gain γ is greater than some threshold value which may depend on the initial conditions.

Let us investigate this case in detail. Recall that we have assumed that the system (26) is semipassive with a radially unbounded storage function $V_1 : \mathbb{R}^n \rightarrow \mathbb{R}^+$. For the sake of simplicity assume that V_1 is differentiable: $V_1 \in \mathcal{C}^1$. Also assume that the function H from the definition of semipassive systems satisfies the *strict* condition: it is strictly positive outside some ball:

$$\exists \rho > 0 |x| \geq \rho \implies H(x) \geq \varrho_1(|x|) \quad (29)$$

where $x = \text{col}(z, y)$ and $\varrho_1 : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is some positive continuous function. Semipassive systems which satisfy (29) will be referred to as *strictly semipassive*.

Since V_1 is radially unbounded it is possible to choose a constant D such that the set $\{x \in \mathbb{R}^n : |x| \leq \rho\}$ belongs to the following compact set $\Omega = \{x \in \mathbb{R}^n : V_1(x) \leq D\}$. If, additionally, the function V_1 is such that outside Ω it satisfies

$$x \in \mathbb{R}^n \setminus \Omega \implies V_1(x) \geq \varrho_2(|x|) \quad (30)$$

where $\varrho_2 : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is a positive continuous non-decreasing radially unbounded function then all solutions of the uncontrolled ($u \equiv 0$) system (26) are ultimately bounded:

$$\overline{\lim}_{t \rightarrow \infty} |z(t)| \leq \mathcal{B}_z, \quad \overline{\lim}_{t \rightarrow \infty} |y(t)| \leq \mathcal{B}_y,$$

where the bounds $\mathcal{B}_z, \mathcal{B}_y$ are such that

$$\Omega \subset \{x \in \mathbb{R}^n : |z| \leq \mathcal{B}_z, |y| \leq \mathcal{B}_y\}$$

and moreover the bounds $\mathcal{B}_z, \mathcal{B}_y$ do not depend on the initial conditions. By virtue of the Yoshizawa

theorem [Yoshizawa, 1960] this result directly follows from the dissipation inequality written for $u \equiv 0$ in the infinitesimal form:

$$\dot{V}_1(x) \leq -H(x).$$

Recall that systems which possess such a property are referred to as ultimately bounded or *dissipative in the sense of Levinson*, or $\mathcal{L}$ -*dissipative*.

It is important to notice that if the system (26) is strictly semipassive with the storage function V_1 then all solutions of the whole system consisting of k coupled systems (25) are ultimately bounded:

$$\overline{\lim}_{t \rightarrow \infty} |z_i(t)| \leq \mathcal{B}_z, \quad \overline{\lim}_{t \rightarrow \infty} |y_i(t)| \leq \mathcal{B}_y, \\ i = 1, \dots, k$$

and the bounds $\mathcal{B}_z, \mathcal{B}_y$ can be chosen *independently* on the initial conditions and on the number of interacting systems k . To prove this fact consider the following scalar function $W : \mathbb{R}^{kn} \rightarrow \mathbb{R}^+$:

$$W(\bar{x}) = \sum_{i=1}^k V_1(x_i)$$

where $\bar{x} = \text{col}(x_1, \dots, x_k)$. Calculating its time derivative in view of the dissipation inequality for semipassive systems and noticing that

$$-y_i^\top y_i + 2y_i^\top y_j - y_j^\top y_j \leq 0$$

we immediately obtain that

$$\dot{W}(\bar{x}) \leq -\sum_{i=1}^k H(x_i)$$

and the result now follows from the Yoshizawa theorem [Yoshizawa, 1960].

Now let us investigate the case of interaction between k strictly semipassive linearly coupled identical systems whose storage functions satisfy all made assumptions. Since all solutions of the whole system are ultimately bounded and in view of the *semigroup property* of dynamical systems without loss of generality we may assume that $|z_i(0)| \leq \mathcal{B}_z$, $|y_i(0)| \leq \mathcal{B}_y$, $i = 1, \dots, k$. Now calculate the time derivative of the function (27):

$$\dot{V}(\bar{z}, \bar{y}) \leq \sum_{i=1}^k \sum_{j=i+1}^k (-\alpha |z_i - z_j|^2 \\ + (C_4 C_1 + C_3) |z_i - z_j| \cdot |y_i - y_j| \\ - (k\gamma \lambda_{\min}(B) - C_2) |y_i - y_j|^2)$$

where the constants $C_1, \dots, C_4$ are chosen to satisfy (22) for all $|z_1| \leq \mathcal{B}_z$, $|z_2| \leq \mathcal{B}_z$, $|y_1| \leq \mathcal{B}_y$, $|y_2| \leq \mathcal{B}_y$ and $\lambda_{\min}(B)$ stands for the minimal eigenvalue of the matrix B , which is positive since B is a positive definite matrix.

Notice that the choice $\gamma > \bar{\gamma}$ where

$$k\bar{\gamma} \geq \frac{1}{\lambda_{\min}(B)} \left(C_2 + \frac{(C_4 C_1 + C_3)^2}{4\alpha} \right)$$

ensures the synchronization goal (28) and since the constants $C_1, \dots, C_4$ and $\mathcal{B}_z, \mathcal{B}_y$ can be chosen independently on k and on the initial conditions one can deduce the following result:

Proposition 1. *The bound $\bar{\gamma}$ of the coupling constant which ensures global synchronization of k identical systems (25) decays as k increases at least as k^{-1} .*

Remark 4. One can notice that the condition of strict semipassivity (or, more mildly, feedback equivalence to a strictly semipassive system) is sufficient but not necessary: indeed the constants $C_1, \dots, C_4$ can be chosen independently on k and on the initial conditions if the right hand side of the whole system is globally Lipschitz and the function V_0 is quadratic but in this case the whole system may have synchronizing unbounded solutions.

The result of Proposition 1 has interesting “biological” interpretations. Suppose we have k living cells whose biochemical processes can be described by the identical systems of ordinary differential equations. Suppose also that these k cells interact such that the model of each of them can be described by the system (25). In this case the condition that the matrix B is positive definite means that the interaction between the cells is of diffusive type and in this case the matrix B and the constant γ are given and cannot be changed. Let the matrix B and numbers k and γ be such that there is no synchrony between the cells then adding perhaps only one cell into the colony such that it begins to interact with the other cells may cause synchronization effects between all cells. Thus examining the oscillatory behavior of interconnected systems and considering the number of systems as a bifurcation parameter one can observe a kind of “synchronizing bifurcation” for some critical number of interacting systems.

For the sake of completeness we should say that this biological interpretation of Proposition 1 was motivated by the paper by S. Smale in [Marsden & McCracken, 1976] who studied a Turing system which is close to (16). He investigated the case when the unconnected systems are globally asymptotically stable but being coupled with the diffusive connection they become oscillatory. An approach to analysis and design of such systems based on the famous Frequency theorem (*Kalman–Yakubovich Lemma*) was proposed in [Tomberg & Yakubovich, 1989] where additionally some problems posed by S. Smale were solved.

6. Synchronization and Convergent Systems

Now let us continue to study the problem of controlled synchronization of two interconnected systems. The case of multiple interconnections can be investigated in the similar way. Notice that according to Assumptions A2, A3 the zero dynamics system (20) has a globally hyperbolically stable solution z_1 (it is allowed that z_1 is not *globally exponentially* stable since Assumption A3 in view of Remark 2 may be satisfied not for all $z_1(t), y_1(t), z_2(t)$). It is interesting whether it is possible to find such a function V_0 whose derivative with respect to the zero dynamics system (20) satisfies Assumption A3 for all continuous bounded functions

$$y_1 \in \mathcal{Y}_B = \{y \in \mathcal{C}^0 : |y(t)| \leq \mathcal{B}_y \forall t \geq 0\}$$

This property can be established by the use of the concept of convergent systems which was introduced by V. A. Pliss and generalized by B. P. Demidovich [Demidovich, 1967]. Motivated by the peculiarity of the synchronization problem we introduce here this concept for the systems of specific form.

Consider the following system:

$$\dot{z} = q(z, d) \quad (31)$$

where $z(t) \in \mathbb{R}^s$, $d(t) \in \mathbb{D}$, $\mathbb{D}$ is some compact subset of $\mathbb{R}^p$, the function $d : \mathbb{R}^1 \rightarrow \mathbb{D}$ is assumed to be continuous: $d \in \mathcal{C}^0$ and the vector field $q : \mathbb{R}^s \times \mathbb{D} \rightarrow \mathbb{R}^s$

is locally Lipschitz continuous in z and continuous in d .

Following B. P. Demidovich [Demidovich, 1967] we give the following definition:

Definition 8. The system (31) is said to be *convergent* if

- (i) all solutions $z(t)$ are well defined for all $t \in \mathbb{R}^1$ and all initial conditions $z(t_0)$.
- (ii) there exists a globally asymptotically stable unique solution $\bar{z}(t)$ bounded for all $-\infty < t < \infty$, i.e. for any solution $z(t)$ it follows that

$$\lim_{t \rightarrow \infty} |z(t) - \bar{z}(t)| = 0$$

If, additionally, system (31) is convergent for all continuous functions d from the given class $\mathcal{D} = \{d \in \mathcal{C}^0 : \mathbb{R}^1 \rightarrow \mathbb{D}\}$ defined by the set $\mathbb{D}$, the system (31) is referred to as *convergent in the class $\mathcal{D}$* .

According to [Demidovich, 1967] there exists a simple sufficient condition which guarantees that the system (31) is convergent (see [Demidovich, 1967], Theorem on p. 286):

Theorem 2. Assume that all eigenvalues of the symmetrized Jacobi matrix

$$J_s = \frac{1}{2} \left[\left(\frac{\partial q}{\partial z}(z, \zeta) \right) + \left(\frac{\partial q}{\partial z}(z, \zeta) \right)^\top \right]$$

are negative² for all $z \in \mathbb{R}^s$ and $\zeta \in \mathbb{D}$.

Then system (31) is convergent in the class $\mathcal{D}$.

Proof of this theorem can be performed by the use of the quadratic Lyapunov function $V = e^\top e/2$, where $e(t) = z(t) - \bar{z}(t)$ and $\bar{z}$ is the unique bounded for all $-\infty < t < \infty$ solution corresponding to the particular function $d \in \mathcal{D}$. The existence of a unique solution $\bar{z}$ can be deduced from the hypothesis of Theorem 2. It can be shown that the conditions of Theorem 2 imply that $\dot{V} \leq -\alpha V$ where $-\alpha$ is the upper bound of the largest eigenvalue of the symmetrized Jacobi matrix, that is the function V satisfies Assumption A3 of Theorem 1.

This theorem allows one to establish a useful connection between the problem of synchronizing output feedback design and the convergency

²Since J_s is a symmetric matrix all its eigenvalues are real numbers

property. Namely, we formulate the following consequence of Theorem 1:

Corollary 1. *Assume that Assumptions A1, A2, A4 of Theorem 1 are satisfied and additionally all eigenvalues of the symmetrized Jacobi matrix*

$$J_s = \frac{1}{2} \left[\left(\frac{\partial q}{\partial z}(z, \zeta) \right) + \left(\frac{\partial q}{\partial z}(z, \zeta) \right)^\top \right]$$

are negative for all $z \in \mathbb{R}^{n-m}$ and $|\zeta| \leq \mathcal{B}_y$ where $\mathcal{B}_y$ is from the proof of Theorem 1. Then for any initial conditions $z_1(0), z_2(0), y_1(0), y_2(0)$ there exists $\bar{\gamma}$ such that for all $\gamma > \bar{\gamma}$ all solutions $z_1(t), z_2(t), y_1(t), y_2(t)$ are bounded for all $t \geq 0$ and the goal of synchronization is achieved.

Now let us summarize the theoretical results of this section. Suppose we have two identical systems of relative degree one globally diffeomorphic to the following normal form:

$$\begin{cases} \dot{z} = q(z, y) \\ \dot{y} = a(z, y) + b(z, y)u \end{cases} \quad (32)$$

Suppose also that for $u \equiv 0$ this system possesses bounded solutions (warning: this condition is necessary but not sufficient to ensure semipassivity). Then if the system

$$\dot{z} = q(z, y)$$

is convergent for all admissible functions $y : \mathbb{R}^+ \rightarrow \mathbb{R}^m$ then two systems of the form (32) being linearly coupled may synchronize for sufficiently large coupling constant. The following example demonstrates how this result can be applied to obtain synchronization of two Lorenz systems.

Example 1. Synchronization of two Lorenz systems, I. Consider the following system:

$$\begin{cases} \dot{x}_1 = \sigma(y_1 - x_1) + u_1 \\ \dot{y}_1 = rx_1 - y_1 - x_1z_1 \\ \dot{z}_1 = -bz_1 + x_1y_1 \end{cases} \quad \begin{cases} \dot{x}_2 = \sigma(y_2 - x_2) + u_2 \\ \dot{y}_2 = rx_2 - y_2 - x_2z_2 \\ \dot{z}_2 = -bz_2 + x_2y_2 \end{cases} \quad (33)$$

with

$$u_1 = -\gamma(x_1 - x_2) \quad u_2 = -\gamma(x_2 - x_1)$$

The problem is to find an appropriate gain γ which ensures boundedness of all solutions and the

following synchronization goal:

$$|x_1(t) - x_2(t)| \rightarrow 0$$

$$|y_1(t) - y_2(t)| \rightarrow 0$$

$$|z_1(t) - z_2(t)| \rightarrow 0$$

as $t \rightarrow \infty$.

First we should check that the system

$$\begin{cases} \dot{x}_1 = \sigma(y_1 - x_1) + u \\ \dot{y}_1 = rx_1 - y_1 - x_1z_1 \\ \dot{z}_1 = -bz_1 + x_1y_1 \end{cases} \quad (34)$$

is $\mathcal{C}^0$ -semipassive with respect to the input u and output x_1 . To this end consider the following smooth scalar function [Neimark & Landa, 1987]:

$$V(x_1, y_1, z_1) = \frac{1}{2}(x_1^2 + y_1^2 + (z_1 - \sigma - r)^2) \quad (35)$$

Its time derivative with respect to the uncontrolled system ($u(t) \equiv 0$) satisfies:

$$\begin{aligned} \dot{V}(x_1, y_1, z_1) &= -\sigma x_1^2 - y_1^2 \\ &\quad - b \left(z_1 - \frac{\sigma + r}{2} \right)^2 + b \frac{(\sigma + r)^2}{4} \end{aligned}$$

It is seen that $\dot{V} = 0$ determines an ellipsoid outside which the derivative of V is negative. If K satisfies

$$K^2 = \frac{1}{4} + \frac{b}{4} \max \left\{ \frac{1}{\sigma}, 1 \right\}$$

then this ellipsoid lies inside the ball

$$\Xi = \{x, y, z : x^2 + y^2 + (z - \sigma - r)^2 \leq K^2(\sigma + r)^2\} \quad (36)$$

that means that all solutions of the uncontrolled system within some finite time approach the set defined by Eq. (36). Additionally, it is obvious that $(\nabla V)^\top g = x_1$. Therefore the function V is a storage function which proves semipassivity of the system (34). Moreover one can notice that the system (34) is strictly semipassive.

Second we should find the *zero dynamics vector field* imposing the external constraint $x_1 = x_2$:

$$\begin{cases} \dot{x}_1 = \sigma(y_1 - x_1) \\ \dot{y}_1 = rx_1 - y_1 - x_1z_1 \\ \dot{z}_1 = -bz_1 + x_1y_1 \\ \dot{y}_2 = rx_1 - y_2 - x_1z_2 \\ \dot{z}_2 = -bz_2 + x_1y_2 \end{cases} \quad (37)$$

Now let us show that $|y_2(t) - y_1(t)| \rightarrow 0$ and $|z_2(t) - z_1(t)| \rightarrow 0$ in (37). It follows from the convergence property of the system

$$\begin{cases} \dot{y}_1 = rx_1 - y_1 - x_1z_1 \\ \dot{z}_1 = -bz_1 + x_1y_1 \end{cases}$$

Indeed, the symmetrized Jacobi matrix for this system has two eigenvalues -1 and $-b$ and therefore, $|y_2(t) - y_1(t)| \rightarrow 0$ and $|z_2(t) - z_1(t)| \rightarrow 0$.

Thus all the conditions of Theorem 1 are satisfied and for any initial conditions there exists a number $\bar{\gamma}$ such that the gain $\gamma > \bar{\gamma}$ ensures boundedness of all solutions and synchronization for all initial conditions (the fact that $\bar{\gamma}$ can be chosen independently on the initial conditions follows from the strict semipassive property of the system (34)).

7. Adaptive Synchronization

Theorem 1 establishes sufficient conditions under which two identical systems synchronize. From the practical point of view this result is not always satisfactory. Indeed, to find $\bar{\gamma}$ the model of the system and perhaps the initial conditions must be known. Thus it is interesting to find an adaptive algorithm which tunes γ until synchronization occurs. Such an algorithm can be easily found. The following result is valid:

Theorem 3 (Gain adaptation). *Assume that the hypothesis of Theorem 1 holds and the synchronization gain is updated by*

$$\begin{aligned} \gamma(t) = & \gamma_0 + \lambda_1(y_1(t) - y_2(t))^\top(y_1(t) - y_2(t)) \\ & + \lambda_2 \int_0^t (y_1(s) - y_2(s))^\top(y_1(s) - y_2(s))ds \end{aligned}$$

where $\lambda_1 \geq 0$, $\lambda_2 > 0$ are some numbers.

Then all solutions of the whole system are bounded and the goal of synchronization is achieved for arbitrary initial conditions $z_i(0)$, $y_i(0)$, γ_0 .

The proof of this fact is straightforward and based on the following Lyapunov-like function:

$$\begin{aligned} W(x_1, x_2, \gamma) = & V_1(x_1) + V_2(x_2) \\ & + \frac{1}{2\lambda_2}(\gamma - \lambda_1(y_1 - y_2)^\top(y_1 - y_2))^2 \end{aligned}$$

Example 2. Synchronization of two Lorenz systems, II. Consider the problem of adaptive synchronization of two Lorenz systems, namely we

want to tune the coupling γ to achieve the goal of synchronization. In order to ensure adaptive synchronization the following adaptation algorithm can be applied:

$$\begin{aligned} \gamma(t) = & \gamma_0 + \lambda_1(x_1(t) - x_2(t))^2 \\ & + \lambda_2 \int_0^t (x_1(s) - x_2(s))^2 ds \end{aligned}$$

where $\lambda_1 \geq 0$, $\lambda_2 > 0$.

We carried out computer simulation to show the synchronization effect between two Lorenz systems. Parameters of simulation were chosen as follows: $\sigma = 10$, $b = 8/3$, $r = 28$, $\lambda_1 = 0$, $\lambda_2 = 0.2$, $x_1(0) = 10$, $y_1(0) = 2$, $z_1(0) = 20$, $x_1(0) = -10$, $y_1(0) = -2$, $z_1(0) = 0$, $\gamma_0 = 0$. Figure 1 shows the transient process in the coupled system. Obviously the goal of synchronization is achieved. Figure 2 shows how the adaptive algorithm changes the synchronization gain γ .

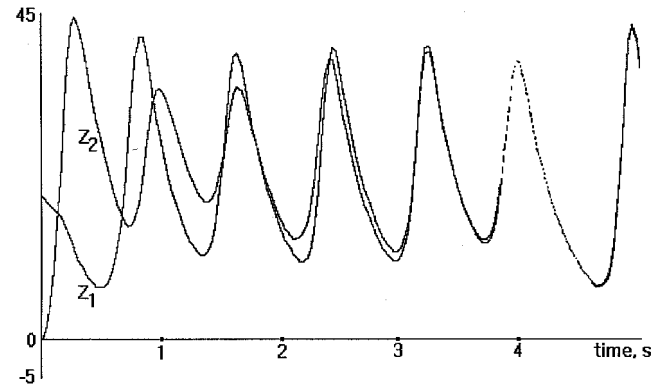


Fig. 1. Synchronization of two Lorenz systems. Comparison of time series obtained from 1st and 2nd systems, $\lambda_2 = 0.2$.

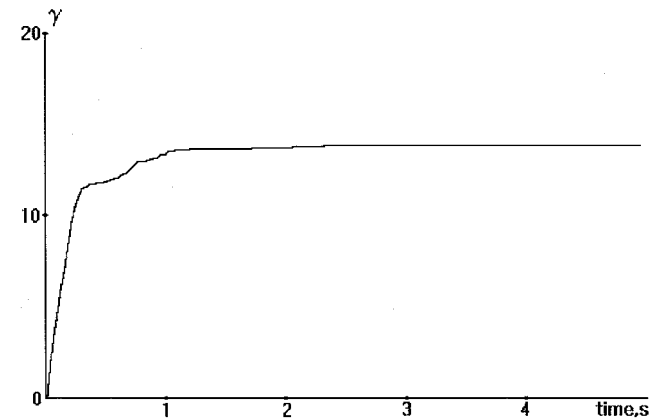


Fig. 2. Synchronization of two Lorenz systems. Adjustment of the synchronization gain γ , $\lambda_2 = 0.2$.

One can see that the synchronization gain increases with time and it tends to some value which depends on the initial conditions and on parameters λ_1, λ_2 . Choosing different parameters λ_1, λ_2 for the same initial conditions one can ensure synchronization for different values of the synchronization gain $\gamma(\infty)$. Figures 3 and 4 show the transient process in the adaptive system corresponding to the parameters $\lambda_1 = 0, \lambda_2 = 0.01$. It is seen that the synchronization gain is smaller compared to the previous figure, but the transient time is greater.

It is important to notice that the system (18) closed by the feedback

$$\begin{cases} u_1 = -\gamma(y_1 - y_2) + v_1 \\ u_2 = -\gamma(y_2 - y_1) + v_2, \quad \gamma > \bar{\gamma} \end{cases} \quad (38)$$

with the new inputs $v_1(t), v_2(t) \in \mathbb{R}^m$ is passive with respect to the input $(v_1 - v_2)$ and output $(b_1(z_1, y_1) + b_2(z_2, y_2))^\top (y_1 - y_2)$. To prove

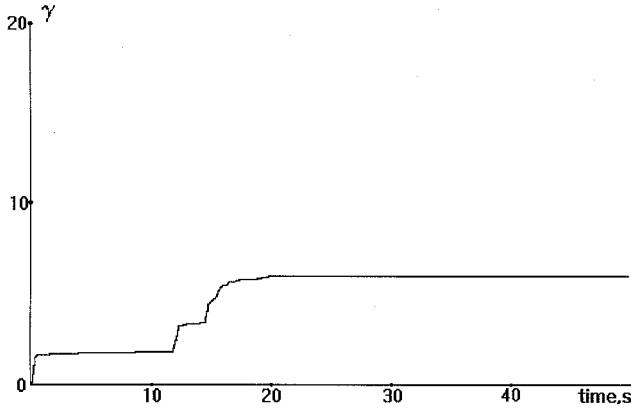


Fig. 3. Synchronization of two Lorenz systems. Adjustment of the synchronization gain γ , $\lambda_2 = 0.01$.

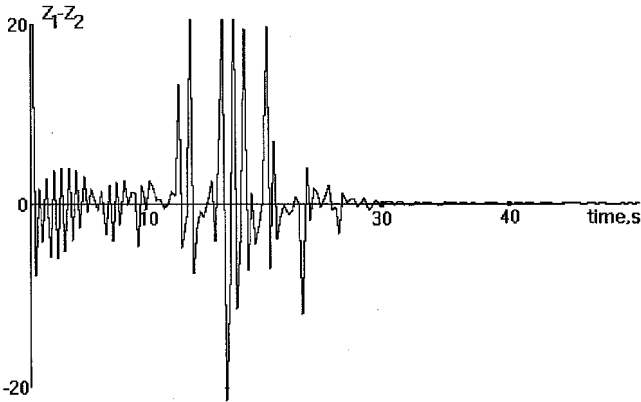


Fig. 4. Synchronization of two Lorenz systems. Error variable $e_z = z_1 - z_2$ versus time, $\lambda_2 = 0.01$.

this fact one can examine the storage function $W_0(z_1, y_1, z_2, y_2) = V(z_1 - z_2, y_1 - y_2)$, where the function V is from the proof of Theorem 1:

$$\begin{aligned} & W_0(z_1(t), y_1(t), z_2(t), y_2(t)) \\ & - W_0(z_1(0), y_1(0), z_2(0), y_2(0)) \\ & \leq \int_0^t ((y_1(s) - y_2(s))^\top (b_1(z_1(s), y_1(s)) \\ & + b_2(z_2(s), y_2(s)))(v_1(s) - v_2(s)) \\ & - \varepsilon(|z_1(s) - z_2(s)|^2 + |y_1(s) - y_2(s)|^2)) ds \end{aligned}$$

Another possible interpretation of the dissipation inequality is very useful for the purposes of adaptive control. Indeed from the dissipation inequality one may conclude that the system is passive with respect to the output $(y_1 - y_2)$ and “virtual” input $\tilde{v} = (b_1(z_1, y_1) + b_2(z_2, y_2))(v_1 - v_2)$. Since the passivity condition as we will see allows one to solve problems of adaptive control it helps to design the desired control $\tilde{v}$ and then if, additionally the matrices b_1 and b_2 depend only on measurable variables one could calculate the desired value of the input $v_1 - v_2$ and then find actual inputs v_1 and v_2 which solve the problem. Moreover, the fact that the synchronization occurs whenever the coupling constant is greater than some threshold value allows for the designer *not to know exactly* the matrices b_1 and b_2 : if the coupling constant is large enough the controller will be *robust* with respect to an uncertainty in b_1 and b_2 .

The property of passivity with respect to the output $y_1 - y_2$ can be reformulated in terms of passivity with respect to sets. Indeed according to the given definitions the feedback (38) makes the system (18) exponentially passive with respect to the set $\mathcal{A}$, where $\mathcal{A}$ is given by

$$\mathcal{A} = \{x_1, x_2 \in \mathbb{R}^n : x_1 = x_2\}$$

This observation will be used in the sequel.

Let us consider the problem of synchronizing output feedback design when the two systems to synchronize differ from each other. Suppose that the high-frequency gains of each systems admit the following factorization:

$$\begin{aligned} b_1(z_1, y_1) &= b'_1(z_1)b''_1(y_1), \\ b_2(z_2, y_2) &= b'_2(z_2)b''_2(y_2) \end{aligned}$$

where $b'_i(z_i) = b'_i(z_i)^\top > 0, i = 1, 2$ are differentiable $m \times m$ positive definite matrix functions and

$b''_i(y_i)$, $i = 1, 2$ are nonsingular matrices. Suppose also that the uncertainty of each systems can be described by the term which acts in span of the inputs $b''_1(y_1)u_1$, $b''_2(y_2)u_2$ (*weakened matching condition* [Fradkov, 1995]). Under this assumption the uncertain overall system can be described by the following equations:

$$\begin{cases} \dot{z}_1 = q(z_1, y_1) \\ \dot{y}_1 = a(z_1, y_1) \\ \quad + b'_1(z_1)(b''_1(y_1)u_1 + r_1(y_1)\theta_1) \\ \dot{z}_2 = q(z_2, y_2) \\ \dot{y}_2 = a(z_2, y_2) \\ \quad + b'_2(z_2)(b''_2(y_2)u_2 + r_2(y_2)\theta_2) \end{cases} \quad (39)$$

where $\theta_1 \in \mathbb{R}^{p_1}$, $\theta_2 \in \mathbb{R}^{p_2}$ are the vectors of unknown parameters, $r_1 : \mathbb{R}^m \rightarrow \mathbb{R}^m \times \mathbb{R}^{p_1}$, $r_2 : \mathbb{R}^m \rightarrow \mathbb{R}^m \times \mathbb{R}^{p_2}$ are known functions of measurable variables. The problem of adaptive synchronization is to find a control law which ensures boundedness of all trajectories of the overall system and achievement of the synchronization goal with respect to the functional

$$\begin{aligned} Q(z_1(\cdot), z_2(\cdot), y_1(\cdot), y_2(\cdot), t) \\ = |z_1(t) - z_2(t)| + |y_1(t) - y_2(t)|. \end{aligned}$$

Obviously it is sufficient to solve this problem under the assumption that $b''_i(y_i) \equiv I_m$. Indeed, if we found the control u_1, u_2 assuming that $b''_i(y_i) \equiv I_m$ then the control $[b''_1(y_1)]^{-1}u_1$, $[b''_2(y_2)]^{-1}u_2$ would solve the problem for arbitrary admissible matrices $b''_i(y_i)$. Motivated by this observation rewrite the system equation in the following form:

$$\begin{cases} \dot{z}_1 = q(z_1, y_1) \\ \dot{y}_1 = a(z_1, y_1) \\ \quad + b'_1(z_1)(u_1 + r_1(y_1)\theta_1) \\ \dot{z}_2 = q(z_2, y_2) \\ \dot{y}_2 = a(z_2, y_2) \\ \quad + b'_2(z_2)(u_2 + r_2(y_2)\theta_2) \end{cases} \quad (40)$$

Solution to the posed problem can be obtained by the following theorem:

Theorem 4. Assume that

A1. The functions q, a, r_1, r_2 are locally Lipschitz

continuous and the functions b'_i , $i = 1, 2$ are differentiable.

A2. The system

$$\begin{cases} \dot{z}_i = q(z_i, y_i) \\ \dot{y}_i = a(z_i, y_i) + b'_i(z_i)(u_i + r_i(y_i)\theta_i) \end{cases} \quad (41)$$

$i = 1, 2$, is $\mathcal{C}^0$ -semipassive with respect to the input u_i and output y_i with the radially unbounded storage function $V_i : \mathbb{R}^n \rightarrow \mathbb{R}^+$.

A3. There exist a $\mathcal{C}^2$ -smooth positive definite function $V_0 : \mathbb{R}^{n-m} \rightarrow \mathbb{R}^+$ and positive number α such that the following inequality is satisfied

$$\begin{aligned} (\nabla V_0(z_1 - z_2))^\top (q(z_1, y_1) - q(z_2, y_1)) \\ \leq -\alpha |z_1 - z_2|^2. \end{aligned}$$

for all $z_1, z_2 \in \mathbb{R}^{n-m}$, $y_1 \in \mathbb{R}^m$.

A4. The matrix $b'_1(z_1) + b'_2(z_2)$ is positive definite:

$$b'_1(z_1) + b'_2(z_2) > 2\beta I_m, \beta > 0.$$

A5. The system (40) is closed by the following adaptive control law

$$\begin{aligned} u_1 &= -\gamma(y_1 - y_2) + v_1 \\ u_2 &= -\gamma(y_2 - y_1) + v_2 \\ v_1 &= (r_2(y_2)\hat{\theta}_2 - r_1(y_1)\hat{\theta}_1)/2 \\ v_2 &= (r_1(y_1)\hat{\theta}_1 - r_2(y_2)\hat{\theta}_2)/2 \\ \hat{\theta}_i(t) &= \psi_i(\sigma_i(y_1(t), y_2(t)) \\ &\quad + \Gamma_i \int_0^t \sigma_i(y_1(s), y_2(s))ds \end{aligned} \quad (42)$$

where

$$\begin{aligned} \sigma_1(y_1, y_2) &= r_1(y_1)^\top (y_1 - y_2) \\ \sigma_2(y_1, y_2) &= r_2(y_2)^\top (y_2 - y_1) \end{aligned}$$

$\Gamma_i = \Gamma_i^\top > 0$ and the functions $\psi_i : \mathbb{R}^{p_i} \rightarrow \mathbb{R}^{p_i}$, $i = 1, 2$ are locally Lipschitz continuous³ and satisfy the pseudogradient condition: $\psi_i(\sigma)^\top \sigma \geq 0 \forall \sigma$.

Then for any initial conditions $z_1(0)$, $z_2(0)$, $y_1(0)$, $y_2(0)$, $\hat{\theta}_1(0)$, $\hat{\theta}_2(0)$ there exists $\bar{\gamma}$

³If the function ψ is discontinuous, e.g. $\psi(\sigma) = \text{sign } \sigma$ then the solution of the overall system should be understood in the sense of Filippov.

such that for all $\gamma > \bar{\gamma}$ all solutions of the overall system are bounded and the following control goal is achieved:

$$|z_1(t) - z_2(t)| \rightarrow 0$$

$$|y_1(t) - y_2(t)| \rightarrow 0 \quad \text{as } t \rightarrow \infty.$$

Proof. First let us prove that in the closed loop system all solutions $x_i = \text{col}(z_i, y_i)$, $\hat{\theta}_i$ are bounded.

To this end consider the following scalar function $W : \mathbb{R}^{2n} \times \mathbb{R}^{p_1} \times \mathbb{R}^{p_2} \rightarrow \mathbb{R}^+$:

$$W(x_1, x_2, \hat{\theta}_1, \hat{\theta}_2) = V_1(x_1) + V_2(x_2) + V_3(\hat{\theta}_1, \hat{\theta}_2)$$

where

$$\begin{aligned} V_3(\hat{\theta}_1, \hat{\theta}_2) &= \frac{1}{4}(\hat{\theta}_1 - \psi_1)^\top \Gamma_1^{-1}(\hat{\theta}_1 - \psi_1) \\ &\quad + \frac{1}{4}(\hat{\theta}_2 - \psi_2)^\top \Gamma_2^{-1}(\hat{\theta}_2 - \psi_2) \end{aligned}$$

It is clear that the function W is radially unbounded. Notice that

$$\begin{aligned} \dot{V}_3 &= \frac{1}{2}(y_1 - y_2)^\top r_1(y_1) \hat{\theta}_1 + \frac{1}{2}(y_2 - y_1)^\top r_2(y_2) \hat{\theta}_2 - \frac{1}{2}(y_1 - y_2)^\top r_1(y_1) \psi_1 - \frac{1}{2}(y_2 - y_1)^\top r_2(y_2) \psi_2 \\ &\leq \frac{1}{2}(y_1 - y_2)^\top r_1(y_1) \hat{\theta}_1 + \frac{1}{2}(y_2 - y_1)^\top r_2(y_2) \hat{\theta}_2 \end{aligned}$$

that is $y_1^\top v_1 + y_2^\top v_2 + \dot{V}_3 \leq 0$. Note also that

$$-y_1^\top y_1 + 2y_1^\top y_2 - y_2^\top y_2 \leq 0.$$

Therefore the function W satisfies the following integral inequality

$$W(x_1(t), x_2(t), \hat{\theta}_1(t), \hat{\theta}_2(t)) - W(x_1(0), x_2(0), \hat{\theta}_1(0), \hat{\theta}_2(0)) \leq \int_0^t (H_1(x_1(s)) + H_2(x_2(s))) ds$$

where the functions H_1 and H_2 are from the definition of semipassive systems. Then as in the proof of Lemma 1 we obtain that all solutions of the overall system are bounded: $|z_1(t)| \leq \mathcal{B}_z, |z_2(t)| \leq \mathcal{B}_z, |y_1(t)| \leq \mathcal{B}_y, |y_2(t)| \leq \mathcal{B}_y$ for some bounds $\mathcal{B}_z, \mathcal{B}_y$ and all time instants $0 \leq t < \infty$.

Then examine the following Lyapunov function candidate:

$$\begin{aligned} V(z_1, z_2, y_1, y_2, \hat{\theta}_1, \hat{\theta}_2) &= V_0(z_1 - z_2) + \frac{1}{2}(y_1 - y_2)^\top (b'_1(z_1) + b'_2(z_2))^{-1}(y_1 - y_2) \\ &\quad + \frac{1}{2}(\hat{\theta}_1 - \psi_1 - \theta_1)^\top \Gamma_1^{-1}(\hat{\theta}_1 - \psi_1 - \theta_1) + \frac{1}{2}(\hat{\theta}_2 - \psi_2 - \theta_2)^\top \Gamma_2^{-1}(\hat{\theta}_2 - \psi_2 - \theta_2) \end{aligned}$$

After simple calculations and applying the pseudogradient condition one can conclude that the time derivative of V satisfies

$$\begin{aligned} \dot{V} &\leq (\nabla V_0(z_1 - z_2))^\top (q(z_1, y_1) - q(z_2, y_1)) + (\nabla V_0(z_1 - z_2))^\top (q(z_2, y_1) - q(z_2, y_2)) \\ &\quad + (y_1 - y_2)^\top (b'_1(z_1) + b'_2(z_2))^{-1}(a(z_1, y_1) - a(z_2, y_2)) - \gamma(y_1 - y_2)^\top (y_1 - y_2) \\ &\quad + \frac{1}{2}(y_1 - y_2)^\top (b'_1(z_1) + b'_2(z_2))^{-1} \frac{\partial b'_1}{\partial z_1}(z_1) \times (b'_1(z_1) + b'_2(z_2))^{-1}(y_1 - y_2) q(z_1, y_1) \\ &\quad + \frac{1}{2}(y_1 - y_2)^\top (b'_1(z_1) + b'_2(z_2))^{-1} \frac{\partial b'_2}{\partial z_2}(z_2) \times (b'_1(z_1) + b'_2(z_2))^{-1}(y_1 - y_2) q(z_2, y_2) \end{aligned}$$

We now aim to determine number $\bar{\gamma} > 0$ such that $\dot{V}$ is nonpositive on the set $|z_1| \leq \mathcal{B}_z, |z_2| \leq \mathcal{B}_z, |y_1| \leq \mathcal{B}_y, |y_2| \leq \mathcal{B}_y$ as $\gamma > \bar{\gamma}$. Using smoothness of the right hand side of (40), invertibility of b'_i ,

$\mathcal{C}^2$ -smoothness of V_0 and boundedness of all solutions $z_1(t)$, $z_2(t)$, $y_1(t)$, $y_2(t)$ we have

$$\begin{aligned} |q(z_2, y_1) - q(z_2, y_2)| &\leq C_1 |y_1 - y_2| \\ |a(z_1, y_1) - a(z_2, y_2)| &\leq C_2 |y_1 - y_2| + C_3 |z_1 - z_2| \\ |\nabla V_0(z_1 - z_2)| &\leq C_4 |z_1 - z_2| \\ \left| \frac{\partial b'_i(z_i)}{\partial z_i} \right| &\leq C_5 \\ |q(z_i, y_i)| &\leq C_6 \\ |(b'_1(z_1) + b'_2(z_2))^{-1}| &\leq C_7 \end{aligned}$$

$i = 1, 2$ for some positive constants $C_1, \dots, C_7$ and all $|z_1| \leq \mathcal{B}_z$, $|z_2| \leq \mathcal{B}_z$, $|y_1| \leq \mathcal{B}_y$, $|y_2| \leq \mathcal{B}_y$. Taking into account Assumption A3, we have

$$\begin{aligned} \dot{V} &\leq -\alpha |z_1 - z_2|^2 \\ &\quad + (C_4 C_1 + C_3) |z_1 - z_2| \cdot |y_1 - y_2| \\ &\quad - (2\gamma\beta - C_2 - C_5 C_6 C_7^2) |y_1 - y_2|^2. \end{aligned}$$

Hence, choosing $\bar{\gamma}$ to satisfy the inequality

$$\bar{\gamma} \geq \frac{1}{2\beta} \left(C_2 + C_5 C_6 C_7^2 + \frac{(C_4 C_1 + C_3)^2}{4\alpha} \right)$$

we just obtain that $\dot{V}$ is nonpositive as long as $|z_1| \leq \mathcal{B}_z$, $|z_2| \leq \mathcal{B}_z$, $|y_1| \leq \mathcal{B}_y$, $|y_2| \leq \mathcal{B}_y$. Then the standard argument completes the proof. ■

It is important to observe that the adaptive control law consists of two independent parts: these are a *passifying* loop and *adaptation* loop. Notice also that the parameter update law can be designed by the so-called speed-gradient algorithm [Fradkov, 1979, 1990; Fradkov & Pogromsky, 1996] with the objective function which coincides with the storage function.

As one can notice the previous theorem gives a semiglobal result. If one looks for a global result one can tune γ like in Theorem 3 or to impose the condition of strict semipassivity instead of semipassivity.

8. Synchronization in the Presence of Disturbances

In the previous sections it was demonstrated how to design an output feedback which globally and/or semiglobally asymptotically synchronizes semipassive systems. The conditions which ensure synchro-

nization also provide feedback passivity of the overall system with respect to the set $\{x_1, x_2 \in \mathbb{R}^n : x_1 = x_2\}$. It is also worth to mention that these conditions together with the *weakened matching condition* allow one to design an adaptive synchronizing output feedback which may be employed to cope with the system uncertainty. Although the presented results are quite promising it is well known that even a very interesting (from the theoretical point of view) control algorithm may be useless in practice because of the disturbances affecting the plant. The problem of how to retain the system performance in the presence of disturbances has been existing since the birth of the control theory and here we do not pretend to give any survey of the possible approaches. Our goal is less ambitious, namely we will show that the approach based on the concepts of passivity is very useful to prevent the systems from the loss of synchrony in the synchronization schemes discussed above.

In the study of this problem we will discuss the control goal of synchronization as a goal of stabilization of some sets. This allows one to avoid the unnecessary details and to divide the whole problem of the synchronization in the presence of disturbances into two independent ones: how to synchronize the systems if there are no disturbances and how to modify the control algorithm to retain the synchrony in the presence of disturbances. Nevertheless we emphasize that asymptotic stability of the “diagonal” set $\{x_1, x_2 \in \mathbb{R}^n : x_1 = x_2\}$ does not imply in general that $\lim_{t \rightarrow \infty} |x_1(t) - x_2(t)| = 0$ (e.g. this diagonal set may possess asymptotically *orbitally* stable solution).

Consider the following problem. Given the system

$$\begin{cases} \dot{x} = f(x) + g_1(x)u + g_2(x)w \\ y = h(x) \end{cases} \quad (43)$$

with the state $x(t) \in \mathbb{R}^n$, input $u(t) \in \mathbb{R}^m$, output $y(t) \in \mathbb{R}^m$ and external disturbance $w(t) \in \mathbb{R}^m$, $f(0) = 0$, it is assumed that f, g_1, g_2, h are $\mathcal{C}^2$ -smooth functions of their arguments. Suppose that the undisturbed and uncontrolled system ($u \equiv 0, w \equiv 0$) possesses an asymptotically stable set $\mathcal{A}$. This set can be understood as a limit set containing a synchronous solution of the overall system. We will assume also that the output of the system which is available for measurements equals to zero for all points in $\mathcal{A}$: $h(x) = 0$ if $x \in \mathcal{A}$. The problem is to find an output feedback which asymptotically stabilizes this set for all initial conditions $x(0)$ from the given compact set Ω and for any admissible disturbance w .

The following assumption restricts the class of admissible plants we shall deal with:

Assumption 1. The system (43) is $\mathcal{C}^1$ -exponentially passive with respect to the set $\mathcal{A}$ with the input $v = u + w$, output y and storage function V .

First let us consider the case when the disturbance $w(t)$ is a bounded function of time such that $\|w\|_\infty \leq \bar{w}$. The following result can be easily derived:

Theorem 5 (High-gain control). *Assume that Assumption 1 is satisfied and the disturbance w is a continuous bounded function of time: $w \in \mathcal{C}^0 \cap \mathcal{L}_\infty$, $\|w\|_\infty = \bar{w}$. Suppose also that boundedness of $|x(t)|_{\mathcal{A}}$ implies boundedness of⁴ $x(t)$.*

Then for any positive number ε , for any initial conditions and for any constant $\bar{w}$ there exists a positive number $\bar{\gamma}$ such that the controller

$$u = -\gamma y$$

ensures realization of the following control objectives

$$\overline{\lim}_{t \rightarrow \infty} |x(t)|_{\mathcal{A}} \leq \varepsilon, \quad \overline{\lim}_{t \rightarrow \infty} V(x(t)) \leq \varepsilon.$$

Proof. Rewrite the dissipation inequality for the function V in the infinitesimal form:

$$\dot{V} \leq -\alpha_3 |x|_{\mathcal{A}}^2 + \bar{w}|y| - \gamma|y|^2$$

It is seen that for sufficiently large γ the inequality $\dot{V} \leq -\delta, \delta > 0$ implies $|y(t)| > \Delta$, that is the latter inequality can be valid only during a finite time interval and the value of $V(x(t))$ is bounded for all $0 \leq t < T_{x_0}$, where T_{x_0} stands for the time limit for which the solution of the closed loop system corresponding to initial conditions $x_0 = x(0)$ exists. Since $|x(t)|_{\mathcal{A}}$ is bounded $x(t)$ is bounded as well and therefore $T_{x_0} = \infty$. Consequently, for sufficiently large γ the value of Δ can be made arbitrary small. Since V is continuous and approaches zero at $|x|_{\mathcal{A}} = 0$ and $h(x) = 0$ when $x \in \mathcal{A}$ the ultimate bound for V can be made arbitrarily small as well by the appropriate choice of γ . ■

This theorem shows that for arbitrary bounded disturbance and initial states it is possible to find such a control gain that all solutions at some time must fall within an ε -distance of the target set $\mathcal{A}$.

It is worth mentioning that the considered feedback does not solve the posed problem: it does not ensure asymptotic stability of the set $\mathcal{A}$. Moreover this algorithm possesses well known drawbacks which are typical for every high-gain controller.

The next result is also not difficult to obtain and it demonstrates how the posed problem can be solved by the variable structure controller.

Theorem 6 (Variable structure feedback). *Assume that Assumption 1 is satisfied and the disturbance w is a continuous bounded function of time: $w \in \mathcal{C}^0 \cap \mathcal{L}_\infty$, $\|w\|_\infty = \bar{w}$. Suppose also that boundedness of $|x(t)|_{\mathcal{A}}$ implies boundedness of $x(t)$.*

Then for any initial conditions the controller⁵

$$u = -\gamma \operatorname{sign} y$$

ensures fulfilment of the goals

$$\lim_{t \rightarrow \infty} |x(t)|_{\mathcal{A}} = 0, \quad \lim_{t \rightarrow \infty} V(x(t)) = 0$$

as long as $\gamma \geq \bar{w}$.

Proof. The result immediately follows from the inequality

$$\dot{V} \leq -\alpha_3 |x|_{\mathcal{A}}^2 - |y|(w + \gamma) \quad (44)$$

⁴Assumptions of this kind are a technical condition which helps to prove boundedness of solutions of the whole system. At first sight it means that $\mathcal{A}$ is compact. But the “diagonal” set $x_1 = x_2$ is not compact and this assumption is satisfied in the case of “master-slave” synchronization and if the solutions of the master system affected by the disturbance are bounded. In more general cases boundedness of solutions can be obtained by the use of the concept of semipassivity similarly to Lemma 1.

⁵The function sign is understood componentwise and the solution to the differential equation with the discontinuous right hand side is understood in the sense of Filippov.

that is $\dot{V}$ is negative definite with respect to $\mathcal{A}$ as long as $\gamma \geq \bar{w}$. Therefore the function V is a non-increasing function of time. Since V is proper with respect to $\mathcal{A}$, $|x(t)|_{\mathcal{A}}$ is bounded for all $0 \leq t < T_{x_0}$. As was assumed boundedness of $|x(t)|_{\mathcal{A}}$ implies boundedness of $x(t)$, hence $T_{x_0} = \infty$. Integrating (44) yields

$$\alpha_3 \int_0^\infty |x(t)|_{\mathcal{A}}^2 \leq V(x(0))$$

The right hand side of the closed loop system is bounded on any bounded set and all solutions are bounded, therefore all solutions of the whole system are uniformly continuous in t . Applying the Barbalat lemma we readily obtain the result. ■

It is easy to notice that all above results remain true if instead of exponential passivity in Assumption 1 one requires strict passivity with the proper with respect to $\mathcal{A}$ storage function.

The VSS-controller studied in the previous theorem solves the problem but it also has some drawbacks typical for VSS systems. Indeed this controller gives a good result if the actuator which realizes the switchings has a fast dynamics, otherwise the controller by itself generates persistent oscillations which are called *chattering*.

Note that we tried to solve the posed problem supposing that the disturbance signal belongs to a very wide class of continuous and bounded functions. At the same time in practice we often face the situation when the disturbance is a periodic or almost periodic function of time and can be presented in the trigonometric series with unknown coefficients:

$$w_i(t) = A_{i0} + \sum_{j=1}^{N_i} A_{ij} \sin(\omega_{ij}t + \phi_{ij}), \quad (45)$$

$$i = 1, \dots, m$$

where the coefficients A_{i0} , A_{ij} , ω_{ij} , ϕ_{ij} are unknown to the system designer.

Suppose that the disturbance w satisfies (45) then it is interesting to find such a *smooth* dynamical output feedback which asymptotically stabilizes the set $\mathcal{A}$. The first step to the solution of this problem can be made using the so called Internal Model Principle (see e.g. [Francis & Wonham, 1975]) which means that the disturbance can be described as an output of some external generator (or exosystem) and to solve the control problem one should con-

sider the augmented plant including the dynamics corresponding to the exosystem.

In [Nikiforov, 1996] there was proposed a state-feedback adaptive compensator of inaccessible disturbance for nonlinear single-input-single-output systems. It was assumed that the model of the plant is known and the whole state vector is accessible to measurement. Under these assumptions the problem of tracking for the reference signal and disturbance cancellation was solved. In [Pogromsky & Nikiforov, 1997] this result was further extended for the multiple-input-multiple-output systems and for the case of incomplete information about the system state. It was shown that exponential passivity of the frozen ($w \equiv 0$) system implies existence of the adaptive compensator which asymptotically semiglobally stabilizes the system at the origin. In this section we reformulate this result focusing to the problem of stabilization of sets. To this end we formulate the main restriction imposed on the disturbance signal.

Assumption 2 (Disturbance model hypothesis).

- (i) The disturbance signal w is bounded and modeled as an output of the following homogeneous system (so-called exosystem)

$$\dot{\chi} = \Gamma \chi \quad (46)$$

$$w = C \chi \quad (47)$$

where $\chi(t) \in \mathbb{R}^d$ is the state vector of the exosystem, a constant $d \times d$ matrix Γ has all its eigenvalues on the imaginary axis and C is a constant matrix. Without loss of generality the pair (Γ, C) is assumed to be observable.

- (ii) The upper bound of the dynamic order d of the exosystem is known, but parameters of the matrices Γ and C are unknown.
- (iii) Neither disturbance w nor the state χ are accessible to measurements.

Additionally to the already made assumptions we need to impose also a condition which restricts the admissible class of the matrix functions g_1, g_2 :

Assumption 3. There exist matrices $G_0 \in \mathbb{R}^{d \times d}$, $K \in \mathbb{R}^{d \times m}$ and vector function $\psi : \mathbb{R}^m \rightarrow \mathbb{R}^d$ such that G_0 is Hurwitz, the pair (G_0, K) is controllable and ψ satisfies the following partial differential equation:

$$\frac{\partial \psi}{\partial y} \frac{\partial h}{\partial x} g_2(x) = K.$$

Let us demonstrate how to verify this assumption. To this end assume that $\text{rank}[(\nabla h)^\top g_2(x)] = m$ for all $x \in \mathbb{R}^n$. It is obvious that if this is the case then there exists a solution $\Psi(x) \in \mathbb{R}^{d \times m}$ to the following linear equation:

$$\Psi(x)(\nabla h)^\top g_2(x) = K.$$

We require this solution to be a known function of measurable variables:

$$\Psi = \bar{\Psi} \circ h$$

where $\bar{\Psi} : \mathbb{R}^m \rightarrow \mathbb{R}^{d \times m}$ is a known matrix function. If, additionally, the following *integrability condition* is fulfilled

$$\frac{\partial \bar{\Psi}_{ik}}{\partial y_j}(y) = \frac{\partial \bar{\Psi}_{ij}}{\partial y_k}(y),$$

$$i = 1, \dots, d, \quad j, k = 1, \dots, m$$

where y_j is the j th entry of the vector y and $\bar{\Psi}_{ik}$ stands for the ik th element of the matrix $\bar{\Psi}$, then there exists a solution $\psi(y)$ to the following partial differential equation

$$\frac{\partial \psi}{\partial y}(y) = \bar{\Psi}(y)$$

and Assumption 3 is satisfied.

Remark 5. It is worth mentioning that Assumption 3 is always satisfied if the matrix g_2 does not depend on the state x : $g_2(x) \equiv \text{const}$.

We need also to restrict the class of admissible matrix functions g_1 :

Assumption 4. The function $\bar{\phi} : \mathbb{R}^n \rightarrow \mathbb{R}^{d \times m}$ is known and depends only on measurable variables: $\bar{\phi} = \phi \circ h$ where $\phi : \mathbb{R}^m \rightarrow \mathbb{R}^{d \times m}$ is a known matrix function and

$$\bar{\phi}(x) = \frac{\partial \psi}{\partial y} \frac{\partial h}{\partial x} g_1(x)$$

Remark 6. As one can see Assumptions 3 and 4 are formulated in the coordinate dependent form, i.e. they can be checked if the system is written in some appropriate coordinate system. Nevertheless in this paper we will not look for the conditions under which the original system is equivalent via output feedback and differentiable coordinate

transformation to the system which satisfies these assumptions.

Now consider the following adaptive controller:

$$\begin{cases} u = -\hat{\theta}\hat{\zeta} - \mu y \\ \dot{\hat{\theta}} = \gamma R(y, \hat{\zeta}) \\ \dot{\eta} = \lambda G_0 \eta + \lambda G_0 \psi(y) - \phi(y)u \\ \dot{\hat{\zeta}} = \eta + \psi(y) \end{cases} \quad (48)$$

where μ is some number, $\hat{\theta} \in \mathbb{R}^{m \times d}$ is the matrix of adjustable parameters, $\gamma, \lambda > 0$ is positive numbers and the $m \times d$ matrix $R(y, \hat{\zeta})$ satisfies

$$R(y, \hat{\zeta}) = (\hat{\zeta}_1 y, \hat{\zeta}_2 y, \dots, \hat{\zeta}_d y), \quad (49)$$

Here $\hat{\zeta}_i$ stands for the i th entry of the vector $\hat{\zeta}$. As one can notice the matrix $R(y, \hat{\zeta})$ is chosen to satisfy the following equation:

$$\text{tr}[R(y, \hat{\zeta})^\top A] = y^\top A \hat{\zeta} \quad (50)$$

for any $m \times d$ matrix A .

The following result generalizes result of [Pogromsky & Nikiforov, 1997] to the case of stabilization of sets:

Theorem 7. Assume that Assumptions 1, 3, 4 are satisfied and for all points in $\mathcal{A}$ it follows that $(\nabla h(x))^\top f(x) = 0$. Suppose also that boundedness of $|x|_{\mathcal{A}}$ implies boundedness of x .

Then for any compact sets $\Omega \subset \mathbb{R}^n$, $\mathcal{U}_1 \subset \mathbb{R}^d$, $\mathcal{U}_2 \subset \mathbb{R}^{m \times d}$ and for any fixed disturbance $w(t)$ satisfying Assumption 2 there exists a positive number $\bar{\lambda} > 0$, such that for any $\lambda \geq \bar{\lambda}$ there exist numbers $\bar{\mu}$ and $\bar{\gamma} > 0$ such that for any initial conditions $x(0) \in \Omega$, $\eta(0) \in \mathcal{U}_1$, $\hat{\theta}(0) \in \mathcal{U}_2$ if $\mu \geq \bar{\mu}$ and $\gamma \geq \bar{\gamma}$ then all solutions of the system (43), (48) are bounded and the following control goals are achieved:

$$\lim_{t \rightarrow \infty} |x(t)|_{\mathcal{A}} = 0, \quad \lim_{t \rightarrow \infty} V(x(t)) = 0$$

Proof. The proof of this Theorem can be performed similarly to the proof of Theorem 1 in [Pogromsky & Nikiforov, 1997] however to make this work expository and self-contained the proof with minor changes is presented here.

First we formulate the following result:

Lemma 3. *Along with (46) and (47) consider the following differential equation*

$$\dot{\zeta} = G\zeta + Kw \quad (51)$$

where $\zeta(t) \in \mathbb{R}^d$ is the state, and the pair (G, K) is controllable. Then for any $d \times d$ Hurwitz matrix G there exists a constant matrix $\theta \in \mathbb{R}^{m \times d}$ such that for a certain initial state $\zeta(0)$ the signal w can be presented in the form

$$w(t) = \theta\zeta(t) \quad (52)$$

The proof of this Lemma is based on the fact that observability of (Γ, C) and controllability of (G, K) guarantee existence of the unique and non-singular solution M to the following Silvester equation [Wonham, 1979] (see also Appendix in the book [Grigoriev *et al.*, 1983]):

$$M\Gamma - GM = KC$$

This equation allows one to find matrix θ and initial conditions $\zeta(0)$ such that (52) is valid: $\theta = CM^{-1}$, $\zeta(0) = M\chi(0)$ (for details see [Pogromsky & Nikiforov, 1997]).

Introduce an auxiliary variable $\delta(t) \in \mathbb{R}^d$ as follows:

$$\delta = \zeta - \hat{\zeta} = \zeta - \eta - \psi(y) \quad (53)$$

Then according to Lemma 3 the disturbance signal can be rewritten in the following form:

$$w(t) = \theta\delta(t) + \theta\hat{\zeta}(t) \quad (54)$$

Calculate the time derivative of δ :

$$\begin{aligned} \dot{\delta} &= \lambda G_0 \zeta + Kw - \lambda G_0 \eta - \lambda G_0 \psi(y) + \phi(y)u \\ &\quad - \frac{\partial \psi}{\partial y} \frac{\partial h}{\partial x}(x)(f(x) + g_1(x)u + g_2(x)w) \\ &= \lambda G_0 \delta - \frac{\partial \psi}{\partial y} \frac{\partial h}{\partial x} f(x) = \lambda G_0 \delta - a(x) \end{aligned}$$

where

$$a(x) = \frac{\partial \psi}{\partial y} \frac{\partial h}{\partial x} f(x)$$

Now rewrite the equation of the overall system:

$$\begin{cases} \dot{x} = f(x) + g_1(x)(-\hat{\theta}\hat{\zeta} - \mu y) + g_2(x)(\theta\delta + \theta\hat{\zeta}) \\ \dot{\delta} = \lambda G_0 \delta - a(x) \\ \dot{\hat{\theta}} = -\gamma R(y, \hat{\zeta}) \end{cases} \quad (55)$$

Notice that $u + w = \hat{\theta}\hat{\zeta} + \theta\delta - \mu y$ where $\hat{\theta} = \theta - \tilde{\theta}$.

Let the positive definite matrix P satisfy the following Lyapunov equation

$$G_0^\top P + PG_0 = -2I$$

Then consider the following Lyapunov function candidate:

$$\begin{aligned} W(x, \delta, \tilde{\theta}) &= V(x) + \frac{\delta^\top P \delta}{2(1 + \delta(0)^\top P \delta(0))} \\ &\quad + \frac{\varepsilon \gamma^{-1} \text{tr}[\tilde{\theta}^\top \tilde{\theta}]}{2(1 + \varepsilon \gamma^{-1} \text{tr}[\tilde{\theta}(0)^\top \tilde{\theta}(0)])} \end{aligned}$$

where the function V (storage function) is from the definition of exponential passivity and the positive number ε is to be determined further.

Fix the set $\bar{\Omega}$ of initial conditions which satisfy

$$\begin{aligned} V(x) + \frac{\delta^\top P \delta}{2(1 + \delta^\top P \delta)} + \frac{\varepsilon \gamma^{-1} \text{tr}[\tilde{\theta}^\top \tilde{\theta}]}{2(1 + \varepsilon \gamma^{-1} \text{tr}[\tilde{\theta}^\top \tilde{\theta}])} \\ \leq 1 + D \end{aligned}$$

for some positive constant D . Consider its projection on the set $\{\delta = 0, \tilde{\theta} = 0\}$. Since V is proper with respect to $\mathcal{A}$ on this projection $|x|_{\mathcal{A}}$ is bounded. Consequently since $x \in \mathcal{A}$ implies $(\nabla h)^\top f(x) = 0$ then due to smoothness of the function a there exists a positive constant L such that, for all $x \in \text{Pr}_{\{\delta=0, \tilde{\theta}=0\}} \bar{\Omega} \subset \Omega_1 \subset \mathbb{R}^n$, where $\Omega_1 = \{x \in \mathbb{R}^n : V(x) \leq 1 + D\}$ we have

$$|a(x)| \leq L|x|_{\mathcal{A}}.$$

It should be also noticed that the constant D may be chosen such that $\Omega \subset \Omega_1$ where Ω is from the theorem statement. In the sequel, we will use the fact that the constant D can be chosen independently on ε , γ and λ , although $\delta(0)$ and $\tilde{\theta}(0)$ depend on λ (it is seen from Lemma 3). Indeed, for any bounded δ and $\tilde{\theta}$ it follows that

$$\frac{\delta^\top P \delta}{2(1 + \delta^\top P \delta)} < \frac{1}{2}$$

$$\text{and} \quad \frac{\varepsilon \gamma^{-1} \text{tr}[\tilde{\theta}^\top \tilde{\theta}]}{2(1 + \varepsilon \gamma^{-1} \text{tr}[\tilde{\theta}^\top \tilde{\theta}])} < \frac{1}{2}$$

and therefore the constant D can determine the upper bound for the initial value of the variable $|x|_{\mathcal{A}}$ (cf. (10)).

Clearly, if $W(x, \delta, \tilde{\theta})$ is bounded then $|x|_{\mathcal{A}}$, δ , $\tilde{\theta}$ are bounded as well. The time derivative of W

along the trajectories of (55) satisfies:

$$\begin{aligned}\dot{W}(x, \delta, \tilde{\theta}) &= -S(x) + y^\top (\tilde{\theta}\hat{\zeta} + \theta\delta - \mu y) - \frac{\delta^\top \lambda \delta}{1 + \delta(0)^\top P \delta(0)} - \frac{\delta^\top P a(x)}{1 + \delta(0)^\top P \delta(0)} - \frac{\varepsilon \operatorname{tr}[R(y, \hat{\zeta})^\top \tilde{\theta}]}{(1 + \varepsilon \gamma^{-1} \operatorname{tr}[\tilde{\theta}(0)^\top \tilde{\theta}(0)])} \\ &\leq -\alpha_3 |x|_{\mathcal{A}}^2 + y^\top \theta \delta - \mu |y|^2 - \frac{\lambda |\delta|^2}{1 + \delta(0)^\top P \delta(0)} - \frac{\delta^\top P a(x)}{1 + \delta(0)^\top P \delta(0)} \\ &\quad - \frac{\varepsilon \operatorname{tr}[R(y, \hat{\zeta})^\top \tilde{\theta}]}{(1 + \varepsilon \gamma^{-1} \operatorname{tr}[\tilde{\theta}(0)^\top \tilde{\theta}(0)])} + y^\top \tilde{\theta} \hat{\zeta}\end{aligned}$$

Determine the numbers $\bar{\lambda} > 0$, $\bar{\mu} > 0$, $\varepsilon > 0$ such that W is positive definite and $\dot{W}$ is nonpositive as long as $\lambda \geq \bar{\lambda}$, $\mu \geq \bar{\mu}$. Let $\bar{\lambda} = \lambda_1 + \lambda_2$, where $\lambda_1 > 0$ is to be chosen and $\lambda_2 > 0$ is fixed, then on the set $\bar{\Omega}$ we have

$$\dot{W}(x, \delta, \tilde{\theta}) \leq W_1(x, \delta, \tilde{\theta}) + W_2(x, \delta, \tilde{\theta}) + W_3(x, \delta, \tilde{\theta})$$

where

$$\begin{aligned}W_1 &= -\alpha_3 |x|_{\mathcal{A}}^2 - \frac{\lambda_1 |\delta|^2}{1 + \delta(0)^\top P \delta(0)} \\ &\quad + \frac{|\delta| \cdot |LP| \cdot |x|_{\mathcal{A}}}{1 + \delta(0)^\top P \delta(0)} \\ W_2 &= -\mu |y|^2 - \frac{\lambda_2 |\delta|^2}{1 + \delta(0)^\top P \delta(0)} + |y| \cdot |\theta| \cdot |\delta| \\ W_3 &= -\frac{\varepsilon \operatorname{tr}[R(y, \hat{\zeta})^\top \tilde{\theta}]}{(1 + \varepsilon \gamma^{-1} \operatorname{tr}[\tilde{\theta}(0)^\top \tilde{\theta}(0)])} + y^\top \tilde{\theta} \hat{\zeta}.\end{aligned}$$

First we observe that if

$$\lambda_1 \geq \frac{|LP|^2}{4\alpha_3} \quad (56)$$

then W_1 is negative definite. It is important that if (56) is satisfied then W_1 is negative definite for all $\delta(0)$ although $\delta(0)$ depends on λ . Further for given λ_1, λ_2 (they determine $\bar{\lambda}$ and therefore $\delta(0)$ and θ) it is possible to find $\bar{\mu}$ such that W_2 is negative definite. Consequently for any $\lambda \geq \bar{\lambda}$, $\mu \geq \bar{\mu}$ it follows that $W_1 + W_2 \leq 0$.

The next step of the proof is to determine the number ε . We have already fixed numbers $\bar{\lambda}$ and $\bar{\mu}$, therefore we determined a compact set of initial conditions $\tilde{\theta}(0)$ and $\delta(0)$ which depends on $\bar{U}_1, \bar{U}_2, \lambda$ and a given disturbance signal $w(t)$ (recall that according to Lemma 3 the matrix θ and initial conditions $\zeta(0)$ depend on the matrix G and hence on λ). Now let us find such ε that $W_3(x, \delta, \tilde{\theta}) = 0$. It is possible if

$$\frac{\varepsilon}{(1 + \varepsilon \gamma^{-1} \operatorname{tr}[\tilde{\theta}(0)^\top \tilde{\theta}(0)])} = 1$$

that is

$$\varepsilon(1 - \gamma^{-1} \operatorname{tr}[\tilde{\theta}(0)^\top \tilde{\theta}(0)]) = 1$$

This equation has a positive solution if the following inequality is satisfied:

$$\gamma \geq \bar{\gamma} = \operatorname{tr}[\tilde{\theta}(0)^\top \tilde{\theta}(0)]$$

Notice that this inequality can be satisfied since we have proved that $\tilde{\theta}(0)$ is bounded and its bound depends on $\lambda, \bar{U}_1, \bar{U}_2$ and λ , in turn, depends on Ω . Additionally recall that the constant D can be chosen independently on ε and γ and therefore for all initial conditions such that $V(x(0)) \leq D$ and arbitrary ε and γ we have for all solutions that $V(x(t)) \leq 1 + D$.

Therefore we have shown that for given initial conditions and a given disturbance from the admissible class it is possible to construct such a positive definite Lyapunov function whose derivative is nonpositive as long as $\lambda \geq \bar{\lambda}$, $\mu \geq \bar{\mu}$, $\gamma \geq \bar{\gamma}$. Therefore W is a nonincreasing function of time. It means that if the initial conditions $x(0)$ belong to the set Ω then $x(t)$ never leaves the set Ω_1 therefore $|x(t)|_{\mathcal{A}}$ is bounded for all $t \geq 0$ and additionally $u(t)$ is bounded as well. As assumed, boundedness of $|x|_{\mathcal{A}}$ implies boundedness of x , therefore all solutions of the overall system are bounded and hence exist on the infinite time interval. From the Barbalat Lemma one can conclude that $W_{1t} \rightarrow 0$ and $W_{2t} \rightarrow 0$ and since $W_3 = 0$ we have that the statement of the theorem holds true for all initial conditions $\{x(0), \eta(0), \tilde{\theta}(0)\}$ from the set $\Omega \times \bar{U}_1 \times \bar{U}_2$ and for the given disturbance signal w . ■

In the next example we will demonstrate how to apply this Theorem to the problem of synchronization of two Lorenz systems one of which is perturbed by the unavailable for measurements periodic disturbance.

Example 3. Synchronization of two Lorenz systems, III. Now let us consider the case of “master-slave” synchronization if the master system (reference model) is affected by an unmeasurable bounded disturbance which satisfies Assumption 2.

Consider the following problem. For the systems

$$\begin{cases} \dot{x}_1 = \sigma(y_1 - x_1) + w \\ \dot{y}_1 = rx_1 - y_1 - x_1z_1 \\ \dot{z}_1 = -bz_1 + x_1y_1 \end{cases} \quad \begin{cases} \dot{x}_2 = \sigma(y_2 - x_2) - u \\ \dot{y}_2 = rx_2 - y_2 - x_2z_2 \\ \dot{z}_2 = -bz_2 + x_2y_2 \end{cases} \quad (57)$$

where $w = w(t) \in \mathbb{R}^1$ is the disturbance, $u = u(t) \in \mathbb{R}^1$ is the control, find such a control which depends only on measurable variables x_1 and x_2 and ensures synchronization goal with respect to the following functional:

$$Q(e_x(\cdot), e_y(\cdot), e_z(\cdot), t) = |e_x(t)| + |e_y(t)| + |e_z(t)|$$

with $e_x = x_1 - x_2$, $e_y = y_1 - y_2$, $e_z = z_1 - z_2$.

We assume that the disturbance signal w satisfies Assumption 2 with $d = 2$. That means that the disturbance w can be presented for example in the following form

$$w(t) = a \sin(\omega t + \varphi)$$

with unknown parameters a , ω , φ .

First it is straightforward to show that for any bounded disturbance w the corresponding solutions $x_1(t)$, $y_1(t)$, $z_1(t)$ are ultimately bounded. Therefore boundedness of $e_x(t)$, $e_y(t)$, $e_z(t)$ implies boundedness of $x_1(t)$, $y_1(t)$, $z_1(t)$, $x_2(t)$, $y_2(t)$, $z_2(t)$.

We will seek the dynamical feedback in the following form $u = -\mu e_x + u_c$ where $-\mu e_x$ is the passifying term and u_c tracks for the inaccessible disturbance (compensating term). Let us find the compensating part of the controller. It can be found in the form

$$u_c = -\hat{\theta}_1 \hat{\zeta}_1 - \hat{\theta}_2 \hat{\zeta}_2$$

where

$$\dot{\hat{\theta}}_i = \gamma \hat{\zeta}_i e_x, \quad i = 1, 2$$

and

$$\hat{\zeta} = \eta + K e_x, \quad \hat{\zeta} = (\hat{\zeta}_1, \hat{\zeta}_2)^\top$$

$$\dot{\eta} = G\eta + GK e_x - K u_c$$

$$G = \begin{bmatrix} -1.5 & 0 \\ 0 & -3 \end{bmatrix}, \quad K = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



Fig. 5. Error e_x versus time. (a) Adaptation is turned off. (b) Adaptation is turned on.

To illustrate the ability of the proposed controller to synchronize two Lorenz systems we carried out computer simulation for the following system parameters: $r = 28$, $b = 8/3$, $\sigma = 10$, $\mu = 20$, $\gamma = 16$. The disturbance signal is simulated as $w(t) = 5 \sin t$. Figure 5(a), presents the error e_x in the system closed by passifying control $u = -\mu e_x$ without compensating one. It is seen that there is an uncompensated error because of the disturbance w . Figure 5(b), shows the time history of e_x in the system closed by the adaptive dynamical compensator. It is seen that the difference $x_1(t) - x_2(t)$ vanishes as time increases.

The considered example has an interesting interpretation. We proposed an algorithm which asymptotically reconstructs the chaotic attractor of the Lorenz system affected by the unknown disturbance. It turns out that using only measured values of the signal x_1 it is possible to estimate the remaining components of the state vector and the disturbance signal w . Nevertheless this controller also has some drawbacks. The main one is that the dynamics of the compensator cannot be made arbitrarily fast for a fixed passifying loop and therefore the transient time cannot be significantly decreased by the appropriate choice of the parameters of the compensator. This disadvantage can be overcome if the whole state vector is available to measurements [Nikiforov, 1996].

9. Conclusions

In this paper the problems of controlled synchronization and adaptive synchronization were considered. We presented some results which give sufficient conditions of existence of a static output feedback which globally and/or semiglobally

asymptotically synchronizes the systems. It should be emphasized that this feedback requires only output on-line measurements, i.e. the synchronization may be achieved under incomplete information about the systems states.

In the paper the proposed approach was also applied in order to investigate synchronization of several identical systems with multiple interconnections. It was shown that in this case the number of interacting systems may play the role of the "bifurcation parameter" at which the systems lose or gain synchronism (Proposition 1).

It has been also shown that the presented results allow one to design an adaptive synchronizing control law which again depends only on output variables and synchronizes the systems with parameter mismatch.

The paper shows that the concepts of strict passivity with respect to sets provide a fruitful design tool for oscillatory synchronizing systems in the presence of disturbances and therefore the main contribution of the paper is that it has been demonstrated that the problems of *controlled synchronization, adaptive synchronization and disturbance rejection can be solved simultaneously and by the use of control which contains a passifying feedback loop*. Indeed, Theorem 1 and Corollary 1 give sufficient conditions for the existence of a static *passifying feedback* and based on the passivity property it is possible to solve the problems of adaptive synchronization (Theorems 3 and 4) and synchronization in the presence of disturbances (Theorems 5–7). Thus the author hopes that the passivity based approach is a promising tool for the design of synchronizing systems.

In this paper we considered the case of static synchronizing output feedback. In future works we will consider a more general case of dynamical output feedback, which allows one to relax some of the conditions imposed on the synchronizing systems.

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References

- Blekhman, I. [1971] *Synchronization of Dynamic Systems* (Moscow-Nauka), in Russian.
- Blekhman, I. [1988] *Synchronization in Science and Technology* (ASME Press, New York).
- Blekhman, I. I., Fradkov, A. L., Nijmeijer, H. & Pogromsky, A. Yu. [1997] "On self-synchronization and controlled synchronization," *Syst. Contr. Lett.* **31**(5), 299–306.
- Byrnes, C. I. & Isidori, A. [1991] "Asymptotic stabilization of minimum-phase systems," *IEEE Trans. Auto. Contr.* **36**(10), 1122–1137.
- Byrnes, C. I., Isidori, A. & Willems, J. C. [1991] "Passivity, feedback equivalence and the global stabilization of minimum phase nonlinear systems," *IEEE Trans. Auto. Contr.* **36**(11), 1228–1240.
- Demidovich, B. P. [1967] *Lectures on Stability Theory* (Nauka-Moscow), in Russian.
- Fradkov, A. L. [1979] "Speed-gradient scheme and its application in adaptive control," *Auto. Remote Contr.* **40**(9), 1333–1342.
- Fradkov, A. L. [1990] *Adaptive Control in Large-Scale Systems* (Nauka-Moscow), in Russian.
- Fradkov, A. L. [1995] "Adaptive synchronization of hyper-minimum-phase systems with nonlinearities," *Proc. 3rd IEEE Mediterranean Symp. New Directions in Contr. Auto.*, Limassol, July 1995, pp. 272–277.
- Fradkov, A. L. & Hill, D. J. [1993] "Exponential feedback passivity and stabilizability of nonlinear systems," Technical report EE9357, Sydney University, Australia.
- Fradkov, A. L. & Pogromsky, A. Yu. [1996] "Speed-gradient control of chaotic continuous-time systems," *IEEE Trans. Circuits Syst. I* **43**(11), 907–914.
- Francis, B. A. & Wonham, W. M. [1975] "The internal model principle for linear multivariable regulators," *Appl. Math. Opt.* **2**, 170–194.
- Grigoriev, V. V., Drozdov, V. N., Lavrentiev, V. V. & Ushakov, A. V. [1983] *Computer Based Design of Discrete Controllers* (Leningrad, Mashinostroenie), in Russian.
- Hill, D. J. & Moylan, P. J. [1976] "The stability of nonlinear dissipative systems," *IEEE Trans. Auto. Contr.* **21**, 708–711.
- Isidori, A. [1989] *Nonlinear Control Systems*, 2nd edition. (Springer-Verlag).
- Kocarev, L., Halle, K. S., Eckert, K., Chua, L. O. & Parlitz, U. [1992] "Experimental demonstration of secure communications via chaotic synchronization," *Int. J. Bifurcation and Chaos* **2**(3), 709–713.
- Lin, Y., Sontag, E. D. & Wang, Y. [1996] "A smooth converse Lyapunov theorem for robust stability," *SIAM J. Contr. Optim.* **34**, 124–160.
- Marsden, J. E. & McCracken, M. [1976] *The Hopf Bifurcation and Its Applications* (Springer-Verlag, New York).

- Morgül, Ö. & Solak, E. [1996] "Observer based synchronization of chaotic systems," *Phys. Rev.* **E54**, 4803–4811.
- Moylan, P. J. [1974] "Implications of passivity in a class of nonlinear systems," *IEEE Trans. Auto. Contr.* Aug. 373–381.
- Neimark, Yu. I. & Landa, P. S. [1987] *Stochastic and Chaotic Oscillations* (Nauka-Moscow).
- Nijmeijer, H. & Mareels, I. [1996] "An observer looks at synchronization," *IEEE Trans. Circuits Syst. I* **44**(10), 882–890.
- Nikiforov, V. O. [1996] "Adaptive nonlinear servocompensation of uncertain external disturbances," *Int. J. Contr.*, to appear.
- Pecora, L. M. & Carroll, T. L. [1990] "Synchronization in chaotic systems," *Phys. Rev. Lett.* **64**, 821–823.
- Pogromsky, A. Yu. & Nikiforov, V. O. [1997] "Output feedback adaptive suppression of forced oscillations in nonlinear systems," *Conf. Informatics and Control, ICI&C'97*, June 9–13, St. Petersburg, Russia.
- Popov, V. M. [1973] *Hyperstability of Control Systems* (Springer-Verlag: New York), (in Romanian: 1966).
- Tomberg, E. A. & Yakubovich, V. A. [1989] "Self-oscillatory conditions in nonlinear systems," *Siberian Math. J.* **30**(4), 180–194, in Russian.
- Willems, J. C. [1972] "Dissipative dynamical systems, Part I: General theory," *Arch. Rational Mech. Anal.* **45**, 321–351.
- Wonham, W. M. [1979] *Linear Multivariable Control: A Geometric Approach* (Springer-Verlag, New York).
- Wu, C. W. & Chua, L. O. [1993] "A simple way to synchronize chaotic systems with application to secure communication systems," *Int. J. Bifurcation and Chaos* **3**(6), 1619–1627.
- Wu, C. W. & Chua, L. O. [1994] "A unified framework for synchronization and control of dynamical systems," *Int. J. Bifurcation and Chaos* **4**(4), 979–998.
- Yoshizawa, T. [1960] "Stability and boundedness of systems," *Arch. Rational Mech. Anal.* **6**, 409–421.